

Playtime–wellbeing associations are practically null across video game genres: a multi-platform digital trace registered report

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Video game research often treats games as a monolithic category, and single-game or single-platform studies leave open whether findings generalise across the medium. In this registered report, we used multi-platform digital trace data from the Open Play dataset to test whether (H1) total playtime is meaningfully associated with mental wellbeing and (H2) whether this association differs by game genre. The data comprise 226,000 hours of playtime telemetry across 4,154 titles from Nintendo, Xbox, and Steam for 1,838 UK and US adults, linked to structured genre metadata and to 6,680 biweekly mental wellbeing assessments (SWEMWBS). Bayesian multilevel within-between models evaluated both hypotheses against a pre-registered threshold for practical significance. Total playtime was practically equivalent to zero on both dimensions: wellbeing barely moved in waves when participants played more than their own average, and those who habitually played more reported slightly lower wellbeing, well under the threshold of practical significance. Contrary to our registered prediction that genres would differ in how playtime relates to fluctuations in mental wellbeing, genre differences fell below the threshold across three model specifications, including a multi-membership model for games carrying multiple genre labels. Genres proved statistically distinguishable but practically interchangeable. A further exploratory analysis extended these tests to the daily timescale and found the same result, namely a small positive within-person trend far below practical significance and a between-person contrast leaning negative within the equivalence region. The playtime–wellbeing null thus survives its strictest test to date and generalises across genres, with slight heterogeneity detected and bounded below any practically meaningful magnitude. Genre differences appear to be the wrong place to look for gaming’s wellbeing effects; genre labels are too coarse a proxy for the psychological experiences that plausibly drive them, which directs attention towards finer-grained characterisations of play.

Keywords: video games, wellbeing, genres, digital trace data, registered report

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Introduction

Video game effects research utilising digital trace data often focuses on a single game or a handful of games to understand the psychological processes and outcomes of digital play (Ballou, Vuorre, et al., 2025; Brühlmann et al., 2020; Johannes et al., 2021; Larrieu et al., 2023). While understandable given the limited resources and lack of objective data available to independent researchers, it has left the field guessing whether identified relationships are unique to the studied titles or whether they apply universally across games. This limitation has restricted researchers’ abilities to estimate the true prevalence of any found effects, leading to assumptions of generalisability and the formation of theories that fail to capture the diversity of the medium. Potential overgeneralisation

of findings has been well-documented in psychological research and is particularly pertinent to research on video games (Yarkoni, 2019).

With more than 10,000 new games released each year (Notis, 2021), an accessible classification system is essential for studying and comparing video games at scale. Genres, while defined by convention and neither exclusive nor exhaustive as labels (Ballou, Hakman, et al., 2025; Clarke et al., 2017; Vargas-Iglesias, 2020), remain the most pragmatic means of categorising video games, providing utility for researchers, players, industry, and policymakers alike. Psychologists have been trying to classify game genres for decades (Griffiths, 1993; Wolf, 2002), but must negotiate challenges in definition, overlaps between genres, and the continuous evolution of genres as new games are released. Genres are therefore best treated not as definitive taxonomical statements but as crude yet practical tools for organising a growing and diversifying library.

Studies have documented marked differences between video game genres in terms of their playtime and their effects on player behaviour, cognition, and wellbeing (André et al., 2024; Dobrowolski et al., 2015; Hazel et al., 2022; Raith et al., 2021). Action games have been linked to increased visual attention (Palaus et al., 2017), cognitive and attentional control (Anguera et al., 2013; Bavelier & Green, 2019), and working memory (Blacker

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This study was pre-registered as part of a Stage 1 Programmatic Registered Report; pre-registration materials are available at <https://osf.io/mvngt/>. The Open Play dataset is openly available on Zenodo at <https://zenodo.org/records/18430947>. All analysis code, including the pre-registered analysis supplement and the multi-membership model validation simulation, is available at <https://github.com/Thomhak/open-play-genres>.

et al., 2014). Massively multiplayer online role-playing games (MMORPGs), first-person shooters, real-time strategy games (RTS), and Multiplayer Online Battle Arena (MOBA) games have been associated with higher gaming disorder (GD) symptoms (Elliott et al., 2012; Rehbein et al., 2021). At the same time, randomised controlled trials have employed genres like casual games and exergames to induce positive moods and reduce stress (Huang et al., 2017; Russoniello et al., 2009). These findings suggest that the effects of video games are not uniform and that genre helps shape both gaming behaviour and its impact on wellbeing.

A recent systematic review, however, documented inconsistencies in how genres are classified across psychological research (Starosta et al., 2024). Of 96 examined articles, only 50 (52%) employed an existing genre classification; 46 (48%) used a unique classification, and 18 (19%) failed to specify the basis or definitions of their genre categorisations altogether. Genre classifications were typically self-reported by participants or assigned by authors, and the same games are habitually classified into different genres across studies. This subjectivity undermines the validity of existing self-report or researcher-ascribed genre taxonomies.

Structured, crowd-sourced metadata repositories such as SteamDB, The Games Database, and the Internet Games Database (IGDB) offer a more reliable alternative, involving both game developers and large player samples in collaboratively maintaining genre labels (Lee et al., 2015; Li & Zhang, 2020; Starosta et al., 2024). This approach accommodates the fluidity of genres over time (Cohen, 1986) and enables comparative research at scale. Yet several methodological obstacles have prevented genre-level research using digital trace data. Data access is fragmented: digital trace data are siloed within individual platform holders, and no cross-platform standard exists for recording or sharing gameplay telemetry (Zendle et al., 2023). The choice of taxonomy is also not neutral: genre labels conflate distinct facets of games, such as gameplay style, purpose, and mood (Lee et al., 2014); treating players as pure-genre versus acknowledging multi-genre play alters conclusions about player characteristics (Azizi et al., 2018), and factor-analytic decompositions of crowd-sourced game tags produce genre structures that diverge from conventional labels (Li, 2020).

The Present Study

In sum, one portion of the literature has shown that playtime undifferentiated by genre is not a meaningful predictor of wellbeing, using data from both specific games (Larrieu et al., 2023) and platforms (Ballou, Sewall, et al., 2024; Ballou, Vuorre, et al., 2025). This relationship, however, has not yet been tested using comprehensive multi-platform play data. Furthermore, multiple mechanisms have been proposed to explain why aggregate playtime is poorly predictive of wellbeing, including displacement of beneficial activities, satisfaction or frustration of basic psychological needs, and selection effects whereby individuals with lower wellbeing play more (Ballou, Hakman, et al., 2025). Most proposed mechanisms operate at the

level of game features or player experience rather than at the level of total time spent; playtime is, at best, a moderator (Ballou, Hakman, et al., 2025). The only causal evidence to date, exploiting console lotteries as a natural experiment, found positive effects of gaming on wellbeing, with heterogeneous effects by console type (Egami et al., 2024), suggesting that observational null findings may reflect confounding rather than absence of effect. Thus, our first hypothesis predicts the absence of a meaningful relationship, this time with digital trace data across several platforms and thousands of games:

H1. Total cross-platform playtime is not meaningfully associated with fluctuations in general mental wellbeing over a 2-week period (**H1a**, “within-person”) or with average wellbeing over the full study period (**H1b**, “between-person”).

We test this via a Bayesian ROPE analysis with equivalence bounds of ± 0.06 SWEMWBS item-mean points per hour/day, matching the SESOI used by Ballou, Sewall, et al. (2024). In practical terms, H1a asks whether playing more than one’s own average in a given two-week wave is associated with a shift in wellbeing, while H1b asks whether individuals who habitually play more report different average wellbeing than those who play less. The within-person component isolates the effect of playtime fluctuations from stable individual differences; the between-person component captures the cross-sectional association, which is inherently more vulnerable to confounding by self-selection.

Consistent null findings for the effects of playtime on wellbeing have generated calls for the unpacking of raw decontextualised playtime into more informative measures that consider games’ varied characteristics and affordances (Ballou, Sewall, et al., 2024). Game genres offer a structured way to differentiate between video games and their corresponding play experiences (Starosta et al., 2024). While genre differences are widely established (Azizi et al., 2018; Dobrowolski et al., 2015; Johnson et al., 2016; Rehbein et al., 2021), the effects of genre-level playtime on wellbeing are yet to be studied using digital trace data. Following previous literature, we predict that genres vary in their relation to wellbeing, with some potentially positive and others potentially negative, varying in strength. Put more formally:

H2. Genres differ in how playtime relates to fluctuations in general mental wellbeing over a 2-week period (**H2a**, “within-person”) and to average wellbeing over the full study period (**H2b**, “between-person”).

H2a asks: when a participant plays more than their own average this wave, does it matter what genres that extra playtime is in? H2b asks the between-person complement: among people who play more overall, does it matter whether that habitual extra playtime is concentrated in RPGs versus Shooters versus other genres? If H2 is supported, and genre-specific associations differ from one another, it would provide a candidate explanation for the consistent null aggregate findings: undifferentiated playtime measures may average over heterogeneous genre effects that cancel in aggregate. H2 thus tests whether it is justified to generalise video game effects across genres,

testing the generalisability that single-title studies must assume (Yarkoni, 2019). Taken together, the two hypotheses form a single generalisability test rather than two separate null results: H1 establishes whether an aggregate association exists, and H2 establishes whether that aggregate conceals genre-level associations that offset one another. The generalisability critique is widely cited but has yet to be operationalised as a quantity a study can estimate.

Beyond the registered hypotheses, we report one unregistered extension. The biweekly window aggregates many individual play sessions into a single retrospective judgement, so the registered tests leave open whether associations operating on shorter timescales are hidden by aggregation. As such, we re-examine both hypotheses at the daily level. This extension was specified after the registered analyses were complete and is reported as exploratory throughout (see Exploratory Daily-Timescale Analysis in Methods).

Methods

Data Source and Participants

The analyses reported here are part of a Stage 1 Registered Report (Ballou, Földes, et al., 2024), and the data collected for this study are part of the Open Play dataset (Ballou, Földes, et al., 2025), a longitudinal dataset that combined multi-platform video game digital trace data alongside daily and biweekly psychological measures from adult gamers in the UK and US over a three-month period. The intake surveys were completed between Sep 2024 and Jan 2026. Digital trace data span the following periods: Nintendo (May 2022 to Oct 2025), Xbox (May 2022 to Sep 2025), Steam (Nov 2024 to Oct 2025), iOS (Mar 2025 to Aug 2025), Android (Mar 2025 to Aug 2025).

This study received ethical approval from the Social Sciences and Humanities Inter-Divisional Research Ethics Committee at the University of Oxford (OII_CIA_23_107). All participants provided informed consent at the start of the study, including consent to their data being shared openly for reanalysis.

Participants were paid at an average rate of £12/hour, equating to: £0.20 for a 1-minute screening, £2 for the 10-minute intake survey (plus £5 for linking at least one account with recent data), £0.80 for each 4-minute daily survey, and £2 for each 10-minute biweekly survey. Participants received a £10 bonus payment for completing at least 24 out of 30 daily surveys and/or 5 out of 6 biweekly surveys.

From 35,369 participants who completed the intake survey, 2,544 met the pre-registered screening and linking criteria. The analytic sample comprises the 1,838 participants who provided valid SWEMWBS responses in at least one biweekly survey wave. Playtime for these participants is measured from their linked Nintendo, Xbox, and Steam accounts; participants and waves without recorded play on these platforms contribute zero hours, and 1,541 participants (84%) had at least one gameplay session of a game that could be matched to a genre. The demographic sensitivity analysis uses the 1,741 participants with a matching qualified intake record and age within the 18–40

screening range (1,041 US, 700 UK; mean age 26.9 years, 66% men); 97 analytic-sample participants are excluded from that subset (97 without a matching intake record, 0 aged over 40). Full demographic characteristics, including country-level breakdowns, are reported in Table A0.1.

Measures

The distributions and structure of the key variables are visualised in Figure 2.

Wellbeing. Wellbeing was assessed using the Short Warwick-Edinburgh Mental Well-being Scale [SWEMWBS; Tennant et al. (2007)], a 7-item short form of the WEMWBS covering psychological functioning and subjective wellbeing over the past 2 weeks, with responses on a 5-point Likert scale ranging from “None of the time” to “All of the time.” Scores are computed as the arithmetic mean of the 7 items (range: 1–5). All reported coefficients, priors, and the pre-registered SESOI are defined on this item-mean scale.

Video game playtime. Digital trace data from Nintendo, Xbox, and Steam provide session-level records (per-title start and end timestamps), while iOS and Android contribute only daily aggregate minutes (no game-level metadata). Nintendo, Xbox, and Steam therefore contribute to both H1 (total playtime) and H2 (genre-specific playtime); iOS and Android contribute only to the pre-registered all-platform sensitivity analysis (Appendix B, Deviation D6). Total playtime is computed from raw sessions without multi-genre duplication (see Multi-genre attribution below).

Game genre classification. Game genre labels for Nintendo and Steam are obtained by cross-referencing game titles in the digital trace data with the Internet Games Database (IGDB). IGDB is, to our knowledge, the only database with complete multi-platform metadata coverage that offers an API for programmatic data retrieval. The genre labels are crowd-sourced: community members can submit contributions (e.g., a new game or alternative categorisation), which are vetted by admins and moderators before appearing in the database. For Xbox, game titles could not be matched to IGDB genre labels, since Xbox provided us only with anonymised title identifiers. Still, genre labels were available through the Xbox store and were mapped onto the IGDB taxonomy. A string-formatting issue in the upstream mapping pipeline had left approximately 67% of Xbox playtime unmapped; we corrected this locally using the `xbox_primary_genre` and `xbox_subgenre` metadata fields, increasing Xbox genre coverage from 33% to 80% of playtime (see Appendix B, Deviation U1). In total, we collected 23 distinct raw IGDB genre labels and collapsed them into 14 study-level categories (e.g., Strategy subsumes Turn-Based Strategy (TBS), Real-Time Strategy (RTS), Tactical, and MOBA; Puzzle subsumes Card & Board Game and Quiz/Trivia). The complete mapping table can be found in Appendix A, and the descriptive statistics by genre in Appendix C. We did not consider IGDB “themes” or other metadata fields. Overall, 94.1% of in-study playtime is matched to at least one study genre (Steam 96.3%, Nintendo 92.3%,

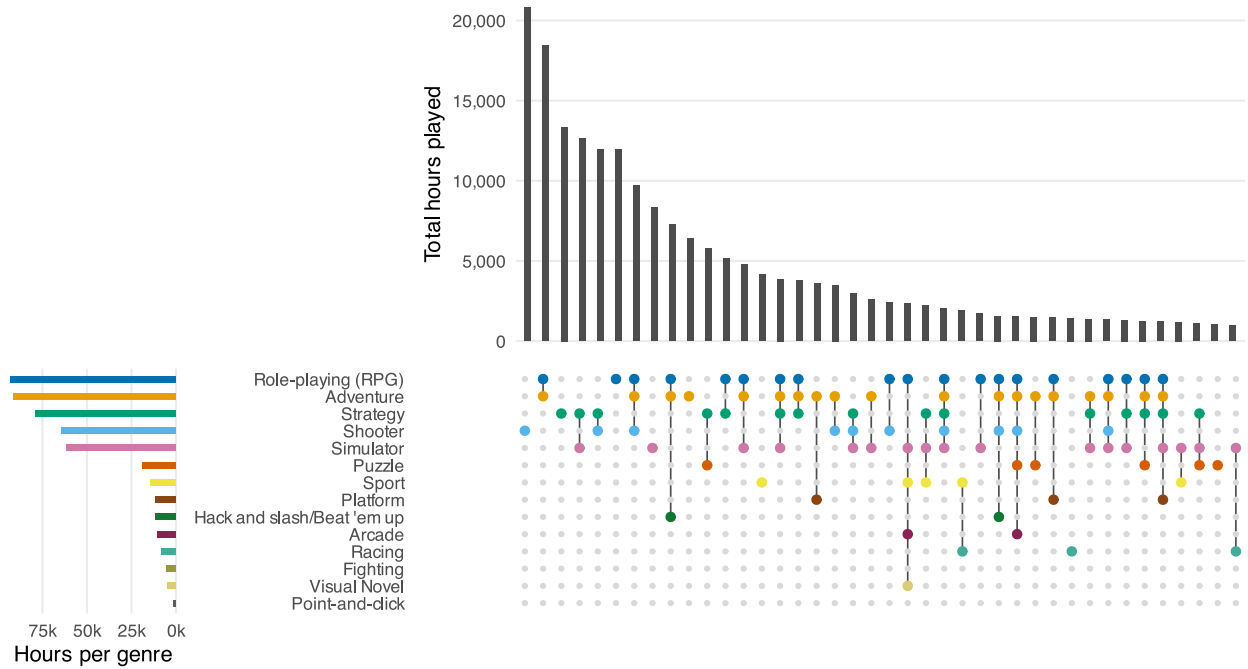


Figure 1. Genre Co-Occurrence Weighted by Total Playtime. Top panel: total hours played by games in each genre combination, ordered by playtime. Bottom panel: dot matrix indicating which genres belong to each combination; connected dots denote multi-genre games. Left bars show total hours attributed to each genre, counting a multi-genre game’s full playtime towards each genre exposure.

Xbox 80.2%); unmatched playtime counts towards total playtime but towards no genre.

Genre co-occurrence and compositional multi-genre attribution. Nowadays, games rarely carry just one genre. Across the 3,721 individual games in our dataset, 62% carry more than one genre label, leading to 325 distinct genre combinations. Figure 1 shows the playtime-weighted co-occurrence structure across the 40 most-played combinations, and Table 1 lists the fifteen most-played combinations with their top titles. Operationalising games as multi-genre has consequential effects in relation to playtime. Namely, instead of treating genre as a partition of time (which only works for single-genre games), we more accurately treat genre as a weighted compositional exposure to that genre’s play: a 60-minute session of an Adventure-RPG (e.g., *Skyrim*) therefore contributes equal weight (i.e., 1) to the Adventure exposure and to the RPG exposure. In the absence of genre weights, partitioning the 60-minute session between the two genres is not only arbitrary (is it 20 minutes of Adventure and 40 minutes of RPG?), it would also understate exposure compared to single-genre games. For example, if that person also played 60 minutes of a single-genre RPG (e.g., *Final Fantasy XIV Online*), the total compositional exposure under a partial attribution of assumed equal weights (30 minutes of Adventure and 30 minutes of RPG for *Skyrim*, and 60 minutes of RPG for *Final Fantasy XIV Online*) would be 0.5 hours of Adventure and 1.5 hours of RPG, which looks like the person played three times as much RPG as Adventure. The genre exposure weights under the full attribution would be 1 hour for Adventure and 2 hours

for RPG, indicating double the RPG exposure compared to Adventure, which reflects the play more accurately than the partial attribution.

Since the playtime of our participants consists of many different multi-genre games played over prolonged periods of time, we operationalise genre exposure as compositional and use full attribution to calculate the genre-exposure weights (see Appendix B, Deviation D1; the pre-registered primary-genre analysis is reported in supplementary materials). As such, each genre coefficient estimates the association between exposure to games carrying that genre and wellbeing.

Demographics. Age, gender, ethnicity, country, and employment were collected at baseline via self-report. Full demographic characteristics are reported in Table A0.1 in Appendix C.

Analytic Approach

All analyses were conducted in R (R Core Team, 2025), and the manuscript was written in Quarto (Allaire et al., 2025). We fit four Bayesian multilevel models with brms (Bürkner, 2017). H1 is addressed by a random-effects within-between (REWB) regression (M0) and by a Bayesian multi-membership model (M3); H2 is addressed by a fixed-effects model (M1), a hierarchical prior model (M2), and the multi-membership model (M3). The three genre models differ in how much information the estimate for one genre shares with the estimates for the others: M1 estimates each genre’s coefficients independently (the pre-registered structure), M2 places a common prior distribu-

Genre combination	Total hours	% of playtime	n games	Mean min/day	Top 5 titles
Shooter	20820	10	268	13	Marvel Rivals [37.2%], Overwatch 2 [11.6%], Call of Duty: WWII [6%], Team Fortress 2 [3.9%], PUBG: Battlegrounds [3.8%]
Adventure + Role-playing RPG	18425	8	213	12	The Elder Scrolls V: Skyrim - Special Edition [8.3%], Clair Obscur: Expedition 33 [6.6%], The Elder Scrolls IV: Oblivion Remastered [5.9%], Elden Ring [5.4%], Throne and Liberty [4.5%]
Strategy	13352	6	139	8	Dead by Daylight [30.9%], R.E.P.O. [16.7%], Dota 2 [11.7%], DeadLock [6.6%], Bloons TD 6 [6.4%]
Simulator + Strategy	12621	6	204	8	RimWorld [18.5%], Schedule I [17%], Europa Universalis IV [6.3%], Hearts of Iron IV [6.3%], Factorio [5.8%]
Shooter + Strategy	11952	5	34	8	Counter-Strike 2 [41.6%], Helldivers 2 [17.5%], Apex Legends [12.7%], 5268BC7F-B83E-4277-9B5A-EF6C2A514F81 [9.5%], Tom Clancy's Rainbow Six Siege X [6.9%]
Role-playing RPG	11942	5	138	8	Final Fantasy XIV Online [24.7%], Elden Ring: Nightreign [22.4%], Lightning Returns: Final Fantasy XIII [6.1%], Star Wars: The Old Republic [4.6%], Xenoblade Chronicles X: Definitive Edition [3.1%]
Adventure + Role-playing RPG + Shooter	9689	4	50	6	Warframe [24.5%], Destiny 2 [15.1%], Cyberpunk 2077 [10.3%], Red Dead Redemption 2 [9%], Rust [7.6%]
Simulator	8337	4	212	5	7DAC9E96-85E7-479B-9622-8A7E73C23FA2 [15.8%], Animal Crossing: New Horizons [11%], Garry's Mod [8.8%], CAC402B8-C0BC-4A0D-AFD3-95E5BAD45B34 [6.6%], House Flipper 2 [4.9%]
Adventure + Hack and slash/Beat 'em up + Role-playing RPG	7293	3	52	5	Path of Exile [27.9%], Monster Hunter Wilds [19.8%], Path of Exile 2 [12.1%], Last Epoch [5.4%], Torchlight: Infinite [5.1%]
Adventure	6381	3	358	4	2089FA57-90DD-4B9A-B9C0-43013F5D6E70 [7.9%], C3A5EDE3-F40F-489E-8E4C-EEAA3BD904C9 [5.6%], F4FF1A75-054D-4451-B506-5AE87E111E2C [5%], BE4D7FFB-533D-4185-8E8A-90CEBFF8858D [4.4%], C1E17D99-3740-44FE-B489-0E8B1D728D6E [4.3%]
Puzzle + Strategy	5779	3	49	4	Balatro [49.3%], Magic: The Gathering Arena [15.3%], Marvel Snap [11.1%], Phasmophobia [8.7%], Mahjong Soul [3.4%]
Role-playing RPG + Strategy	5139	2	48	3	Baldur's Gate III [76.9%], Reverse: 1999 [5.6%], Summoners War: Rush [4%], Total War: Rome II - Emperor Edition [3.1%], Digimon Story Cyber Sleuth: Complete Edition [2.1%]
Adventure + Role-playing RPG + Simulator	4763	2	49	3	Palia [25.4%], Kingdom Come: Deliverance II [14.8%], Shop Titans [11.4%], Hello Kitty Island Adventure [11%], Fantasy Life i: The Girl Who Steals Time [7.2%]
Sport	4164	2	60	3	C3C63F46-0627-4660-AAF5-974831379998 [15.5%], BA2D9D35-CA26-4220-9D7D-C41E318FA5CE [14.3%], Rematch [12.4%], C185D171-636E-46E4-B0FE-CB24CB53568B [10%], 97F40174-3110-4CE3-8F42-FE8DC56ED62C [8.8%]
Adventure + Role-playing RPG + Simulator + Strategy	3881	2	33	2	Stardew Valley [72.7%], Mount & Blade II: Bannerlord [14.4%], Coral Island [2.3%], Travellers Rest [2.3%], Caves of Qud [2.3%]

Table 1. Top 15 genre combinations by total playtime. Multi-genre rows join labels with ‘ + ‘. Total hours = playtime for games with that combination; % of playtime = share of all genre-matched playtime; n games = unique game-platform pairs. Xbox titles anonymised.

tion over the genre coefficients and estimates its standard deviation (partial pooling), and M3 assigns each wave to the genres it contains in proportion to their playtime shares (multi-membership). Deviations from the pre-registered

plan are catalogued in Appendix B and are flagged below where they affect interpretation.

All models use a within-between decomposition of playtime: the between-person component is defined as the participant's average across the study, and the within-

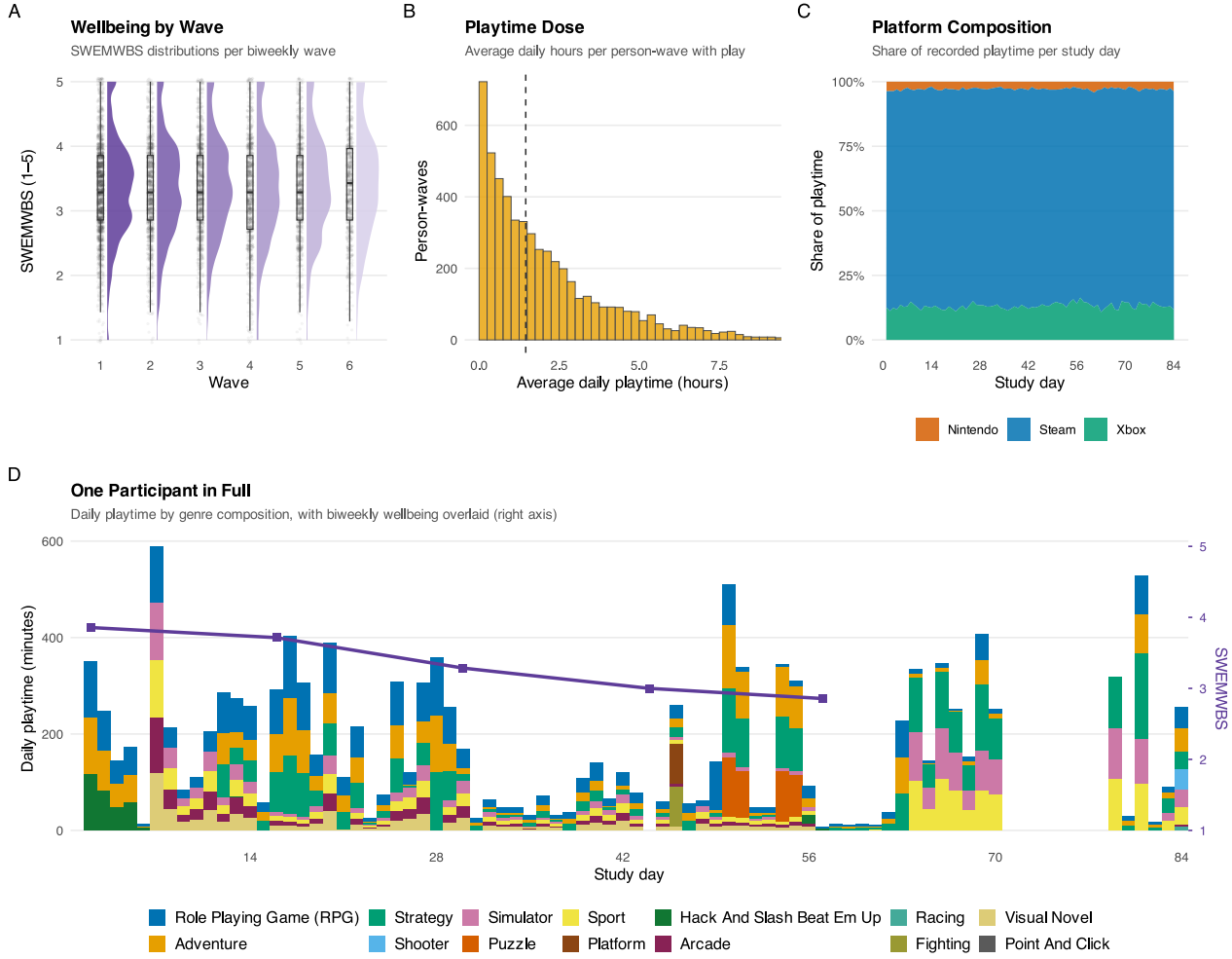


Figure 2. Distributions, Composition, and Raw Structure of the Study Measures. (A) SWEMWBS by wave. (B) Average daily playtime across person-waves with recorded play (15-minute bins; dashed line: median; 19% of person-waves have no recorded play). (C) Platform share of recorded playtime across study days 1–84. (D) Raw measures for one participant at the 75th percentile of playtime (ID p5271179686): daily playtime coloured by genre, with multi-genre time split equally across labels so bar heights are clock time (genre exposures use full attribution). Biweekly SWEMWBS is the purple line, read against the right axis.

person component as the deviation of a given wave from that personal average. The two components answer different questions. The within-person coefficient describes whether a participant’s wellbeing changes in waves in which they play more than they usually do, because each participant serves as their own comparison. The between-person coefficient describes whether participants who habitually play more report different average wellbeing than participants who habitually play less; this comparison is made across people.

M0: Total Playtime

M0 addresses H1: is total playtime across Nintendo, Xbox, and Steam associated with wellbeing? The model is a random-effects within-between (REWB) regression and returns the two coefficients, β_W for within-person fluctuations and β_B for habitual between-person differences. Each participant receives a random intercept, because people differ in baseline wellbeing, and a random slope

for within-person playtime, because the association need not be identical for everyone. Priors are calibrated to the 1–5 outcome scale. The intercept prior is centred on the scale midpoint, and the coefficient priors are weakly informative, leaving effects up to roughly 1 point per hour/day plausible while regularising implausibly large values.

$$\begin{aligned}
 \text{SWEMWBS}_{ij} &\sim \text{Normal}(\mu_{ij}, \sigma) \\
 \mu_{ij} &= \alpha + \beta_W \text{Time}_{\text{within},ij} + \beta_B \text{Time}_{\text{between},j} + u_{0j} + u_{1j} \text{Time}_{\text{within},ij} \\
 \begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} &\sim \text{Normal}(\mathbf{0}, \Sigma) \\
 \alpha &\sim \text{Student-}t(3, 3, 2) \\
 \beta_W, \beta_B &\sim \text{Student-}t(3, 0, 0.5)
 \end{aligned} \tag{1}$$

M1: Fixed-Effects

M1 is the pre-registered specification for H2. Each of the 14 genres receives its own within-person and between-person coefficient, estimated independently, with a random intercept per participant. Independent estimation

has a benefit and a cost. The benefit is that the model imposes no assumptions about how similar genres are to one another. The cost is that rarely played genres are estimated from little data and therefore receive wide credible intervals, which is the problem that M2 and M3 address. Because genre exposures are non-exclusive, they cannot be summed to recover total playtime, and M1 therefore contains no separate total-playtime term. Two departures from the registered specification should be noted: exposures use full attribution rather than primary-genre labels (Deviation D1), and the registered random slopes for all genre terms (28 additional parameters) proved near-singular, so we followed the pre-registered contingency plan and retained the random intercept only.

$$\text{SWEMWBS}_{ij} \sim \text{Normal}(\mu_{ij}, \sigma)$$

$$\begin{aligned} \mu_{ij} &= \alpha + \sum_{g=1}^G (\beta_{W,g} \text{Time}_{\text{within},g,ij} + \beta_{B,g} \text{Time}_{\text{between},g,j}) + u_{0j} \\ u_{0j} &\sim \text{Normal}(0, \tau_{\text{pid}}) \\ \alpha &\sim \text{Student-}t(3, 3, 2) \\ \beta_{W,g}, \beta_{B,g} &\sim \text{Student-}t(3, 0, 0.5) \end{aligned} \quad (2)$$

M2: Hierarchical Prior

M2 keeps M1's structure but changes how the genre coefficients are estimated, and this change yields a direct answer to H2. Rather than treating the 14 coefficients as unrelated, M2 assumes that they are drawn from a common normal distribution centred at zero and estimates the standard deviation of that distribution, τ_{genre} , separately for the within-person and between-person dimensions. τ_{genre} is therefore an estimate of exactly the quantity H2 asks about: how much the genre coefficients differ from one another, expressed in outcome units per hour of play. The shared distribution also produces partial pooling. Estimates for rarely played genres are pulled towards the common centre, whereas genres with ample data retain estimates close to their independent values.

The Half-Normal(0, 0.1) hyperprior on τ is matched to the outcome scale. Genre coefficients are expressed in SWEMWBS points per hour/day, and the smallest effect size of interest is 0.06 on that scale, so a much wider hyperprior would place substantial prior mass on heterogeneity far larger than any effect considered plausible. The sensitivity of the results to halving and doubling this scale is examined in Appendix D. Sampling near $\tau = 0$ is numerically difficult, so we use conservative sampler settings and report divergence counts and R-hat diagnostics for every model alongside the results.

$$\text{SWEMWBS}_{ij} \sim \text{Normal}(\mu_{ij}, \sigma)$$

$$\begin{aligned} \mu_{ij} &= \alpha + \sum_{g=1}^G (\beta_{W,g} \text{Time}_{\text{within},g,ij} + \beta_{B,g} \text{Time}_{\text{between},g,j}) + u_{0j} \\ u_{0j} &\sim \text{Normal}(0, \tau_{\text{pid}}) \\ \beta_{W,g} &\sim \text{Normal}(0, \tau_{\text{within}}) \\ \beta_{B,g} &\sim \text{Normal}(0, \tau_{\text{between}}) \\ \tau_{\text{within}}, \tau_{\text{between}} &\sim \text{Half-Normal}(0, 0.1) \\ \alpha &\sim \text{Student-}t(3, 3, 2) \end{aligned} \quad (3)$$

M3: Multi-Membership

M3 represents each wave the way panel D of Figure 2 displays it, namely as a total amount of playtime with a

genre composition. The model retains total playtime as the predictor, inheriting M0's structure including the overall β_W and β_B , and allows the composition of that playtime to modify the playtime slope. Formally, this is a multi-membership structure, where each wave carries all 14 genre labels and only the weights vary, equal to each genre's share of exposure. As such, a wave consisting mostly of Shooter play is governed mostly by the Shooter slope adjustment, and a wave divided between Adventure and RPG draws on both adjustments in proportion.

Each genre's slope adjustment is drawn from a common distribution, and the standard deviation of that distribution, σ_s , is this model's answer to H2: it quantifies how much the playtime–wellbeing slope varies with the genres being played.

$$\text{SWEMWBS}_{ij} \sim \text{Normal}(\mu_{ij}, \sigma)$$

$$\begin{aligned} \mu_{ij} &= \alpha + \beta_W \text{Time}_{\text{within},ij} + \beta_B \text{Time}_{\text{between},j} + u_{0j} + u_{1j} \text{Time}_{\text{within},ij} \\ &\quad + \sum_{g=1}^G w_{g,ij} (s_{W,g} \text{Time}_{\text{within},ij} + s_{B,g} \text{Time}_{\text{between},j}) \\ \begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} &\sim \text{Normal}(0, \Sigma_u) \\ s_{W,g} &\sim \text{Normal}(0, \sigma_s) \\ s_{B,g} &\sim \text{Normal}(0, \sigma_{s, \text{between}}) \\ \alpha &\sim \text{Student-}t(3, 3, 2) \\ \beta_W, \beta_B &\sim \text{Student-}t(3, 0, 0.5) \\ \sigma_s, \sigma_{s, \text{between}} &\sim \text{Half-Normal}(0, 0.3) \end{aligned} \quad (4)$$

Because this use of multi-membership is unconventional, we validated the specification by parameter recovery before fitting it to real data. Synthetic datasets with known genre-slope heterogeneity were fit with identical model code; every population-level parameter, including σ_s , was recovered within its 95% credible interval, with no divergent transitions (simulation report in the supplementary materials).

Missing Data

Missingness in the outcome takes two forms. Item-level missingness is rare and all-or-nothing: 43 submitted surveys lack SWEMWBS, each missing all seven items. Wave nonresponse is the dominant form: survey completion declined from 100% at wave 1 to 40% at wave 6, leaving 4,348 of the 11,028 analytic person-waves (39.4%) without an observed SWEMWBS. Playtime was not imputed: tracking was continuous and automatic, so within linked platforms an absence of play is recorded absence rather than missing data, and exposures exist for every person-wave.

The primary Bayesian models are fit to the observed outcomes; this is valid provided outcome missingness is at random given the modelled covariates and random effects. To test that assumption, we multiply imputed SWEMWBS on the complete participant $\times$ wave grid, using the two-level protocol shared with the programmatic registered report's sibling Stage 2 studies: multiple imputation by chained equations [mice and micedds; van Buuren & Groothuis-Oudshoorn (2011); Robitzsch & Grund (2024)] with predictive mean matching (21.pmm), participants as random-intercept clusters (van Buuren, 2018), and $m =$

100 imputations set from the pilot fraction of missing information by von Hippel's (2020) two-stage rule. Predictive mean matching donates observed values, so imputed scores respect the 1–5 scale. Frequentist counterparts of the analysis models were fit to each imputed dataset and pooled by Rubin's rules; the pooled results do not meaningfully differ from the complete-case analyses (Appendix D, Table A0.4).

Smallest Effect Size of Interest

Establishing the absence of a practically significant effect requires committing in advance to a threshold below which an association is too small to matter, the smallest effect size of interest (SESOI). By practically significant, we mean the smallest change in wellbeing that a player would themselves notice, known also as a minimally important difference (MID). Following previous studies, the MID is set at 0.3 points on the 1–5 SWEMWBS scale, converging with anchor-based estimates of the WEMWBS [Warwick-Edinburgh Mental Well-being Scale; Ballou, Sewall, et al. (2024); Maheswaran et al. (2012)] and the PANAS [Positive and Negative Affect Schedule; Anvari & Lakens (2021)].

To calculate the SESOI, we also need to know how large a change in playtime is realistically available to reach the MID. Because UK and US adults average approximately 5 hours of daily leisure time (Sturm & Cohen, 2019), this provides us with an upper bound on how much daily play could plausibly rise or fall without it displacing other behaviours that arguably impact wellbeing more than play (e.g., sleep). SESOI is the ratio of the two: $0.3 \div 5 = 0.06$ points per hour. An association smaller than this means that even a person who shifted their entire 5-hour daily leisure time budget into or out of gaming would not experience a noticeable change in wellbeing.

We evaluate all coefficients against the interval between the negative and positive SESOI $[-0.06, +0.06]$, also known as the Region of Practical Equivalence [ROPE; Kruschke (2018)]. If a coefficient's 95% credible interval (CrI) falls entirely inside the ROPE, the effect is considered practically null, meaning even the largest plausible effect is too small to matter; entirely above $+0.06$, practically positive; entirely below -0.06 , practically negative. A CrI straddling either edge of the ROPE is inconclusive.

Exploratory Daily-Timescale Analysis

To assess whether our findings hold on a shorter timescale, we conducted an exploratory daily-timescale analysis. The daily extension is unregistered; an analysis plan fixed the exposure window, sample criteria, model set, priors, and SESOI before any exposure–outcome model was estimated. US participants were asked to complete daily surveys for the first 30 study days, including a life-satisfaction item ("I was satisfied with my life today") answered on a 0–100 slider (Ballou, Földes, et al., 2025), which serves as the outcome. Exposure is console and PC playtime from 4:00 a.m. local time of the response's waking day up to the survey completion timestamp, computed from the same session-level telemetry and full-attribution genre assignment as the main analysis, with local time taken

from each participant's reported timezone; Xbox sessions recorded with date-only timestamps (8% of sessions) have their minutes spread uniformly over the recorded day. Of 15.8K diary responses from 1.2K participants, 13.0K responses (82.6%) from 763 participants had available telemetry and analysable same-day windows; exclusions were dominated by days without telemetry coverage. Excluded responses enter an all-responses sensitivity refit that retains every response and scores unobserved play as zero.

The within-between decomposition is hybrid: within-person terms are deviations from a person's own same-day average, and between-person terms are habitual full-day hours (a participant's total telemetry over their observed diary period divided by days), which keeps habitual contrasts in daily-hour units directly comparable with the registered analysis. Because the pre-registered SESOI is defined on the SWEMWBS item-mean scale, a SESOI for the slider is constructed the same way, dividing a minimally important difference by the roughly five hours of daily leisure realistically available (Sturm & Cohen, 2019). We adopt a minimally important difference of 5 slider points, below both a distribution-based half within-person standard deviation (6.3 points) and the registered minimally important difference expressed as a share of its scale range (7.5%, or 7.5 slider points); the daily SESOI is therefore 1 slider point per hour of daily play, a deliberately conservative choice, since a smaller minimally important difference makes practical equivalence harder to claim rather than easier. The model specifications (M0–M3) are refit at the daily level. Robustness refits (autocorrelated residuals with a day trend, adjustment for window length, exclusion of the date-only Xbox sessions, and the all-responses sample) are reported in Appendix D.

Results

Our analytic sample comprised 1,838 participants with valid SWEMWBS responses in at least one biweekly wave, yielding 6,680 person-wave observations ($M = 3.6$ of 6 waves per participant; survey response declined from 100% at wave 1 to 40% at wave 6, see Missing Data and Appendix D). Playtime is measured from linked Nintendo, Xbox, and Steam telemetry, with zero hours recorded for participants and waves without play on these platforms. Participants played an average of 1.67 hours per day ($SD = 2$) on Nintendo, Xbox, and Steam combined. The most played genres were Role Playing Game (RPG) and Adventure.

Data Quality Controls

Prior to hypothesis testing, we conducted the three pre-registered positive data quality controls (DQCs) to validate expected patterns in the data. First, self-reported biweekly playtime correlated positively with digital trace playtime ($r = 0.469$, one-sided 95% lower bound $= 0.451$, $p < .001$). Second, session integrity largely held. Xbox contained 0 overlapping sessions, and no Steam session's recorded playtime exceeded its elapsed wall-clock duration (0 violations). Nintendo retained a small number of overlapping sessions: 675 of 288,082 sessions (0.23%), with a median

overlap of 9.8 minutes and accounting for 0.09% of total Nintendo playtime. These pairs reflect concurrent use of multiple consoles linked to one account: 529 of the 675 pairs involved two different games recorded as running at once, and 95 involved two distinct console models (e.g., a Switch Lite overlapping an OLED model), which requires two physical devices; because the device identifier records the model rather than the individual console, some same-model pairs likely also reflect two consoles. The overlaps were also concentrated in a few accounts, with two participants alone contributing 236 pairs. Given their rarity and their small share of playtime, we judged these overlaps too small to materially influence the wave-level aggregates. Third, men played more shooter games than women ($M = 0.487$ vs. 0.309 hrs/day, $t(1144.4) = 4.25$, $p < .001$, Cohen's $d = 0.21$), meeting the pre-registered criterion of a significant difference in the expected direction and consistent with population studies of gendered genre preferences (Rehbein et al., 2016). We therefore proceeded with the confirmatory analyses.

H1: Total Playtime and Wellbeing

Figure 3 summarises the H1 results; the underlying posterior summaries for all hypothesis tests are tabulated in Appendix E (Table A0.1). For the within-person effect (H1a), the posterior mean was 0.003 SWEMWBS item-mean points per hour/day (95% CrI: $[-0.008, 0.014]$), with 100% of the posterior inside the ROPE, above the 95% posterior-mass criterion. Therefore, when a participant plays more than their own average in a given wave, their wellbeing does not change to any practically meaningful degree, supporting H1a. For the between-person effect (H1b), the posterior mean was -0.017 (95% CrI: $[-0.035, 0.001]$), with 100% of the posterior inside the ROPE, likewise establishing practical equivalence. The between-person estimate leans negative (its credible interval extends just to zero), but its magnitude is far below the SESOI: even a 2 hour/day difference in habitual playtime

would correspond to only -0.034 points on the 1–5 scale (about 0.8% of the 4-point range).

Both components of H1 are therefore supported. The model also bounds person-to-person variation in the within-person association: the random-slope SD was 0.035 (95% CrI: $[0.005, 0.062]$), with 3.8% of its posterior above the SESOI.

The multi-membership model also provides H1 estimates via its fixed effects. Figure 3 confirms that M0 and M3 yield consistent H1 estimates, and that the inclusion of genre-level terms in M3 does not alter the overall playtime–wellbeing association.

H2: Genre-Specific Effects

H2 asks whether genres differ in how playtime relates to wellbeing. We answer it in two steps. First, we report the genre heterogeneity parameters, which provide a direct test of H2: how much do genre-specific playtime slopes vary? Second, we examine the per-genre estimates the three specifications produce. A further exploratory analysis asks whether these effects hold at the daily level.

Genre Heterogeneity

Figure 4 shows the posterior distributions for the genre heterogeneity statistics from all three H2 models. M1's statistic is the standard deviation of the 14 independent within-person genre coefficients; M2's is τ_{genre} from the learned hierarchical prior; M3's is σ_s from the multi-membership random effects. All three are in the same units (SWEMWBS item-mean per hour/day) and represent how much genre-specific playtime slopes vary across genres.

The primary test of H2 comes from the multi-membership model's (M3) genre heterogeneity parameters. For the within-person dimension (H2a), the posterior mean of $\sigma_{s, \text{within}}$ was 0.044 (95% CrI: $[0.006, 0.1]$). Because a standard deviation cannot be negative, the ROPE is the SESOI of 0.06, which is exceeded by 21.9% of the posterior. The data are thus most consistent with genre-to-genre variation in within-person slopes that is smaller than the

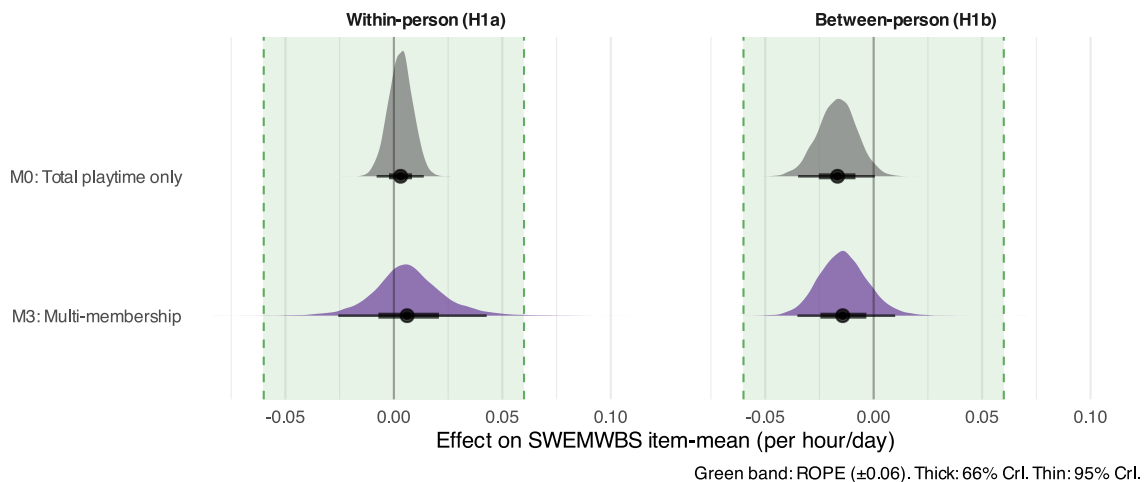


Figure 3. Posterior Distributions for Total Playtime Effects (H1). M0 is the standalone H1 model; M3 adds genre information through multi-membership weights while still estimating overall playtime effects. M1 and M2 have no separate total-playtime predictor and are excluded. Green band: pre-registered ROPE (± 0.06).

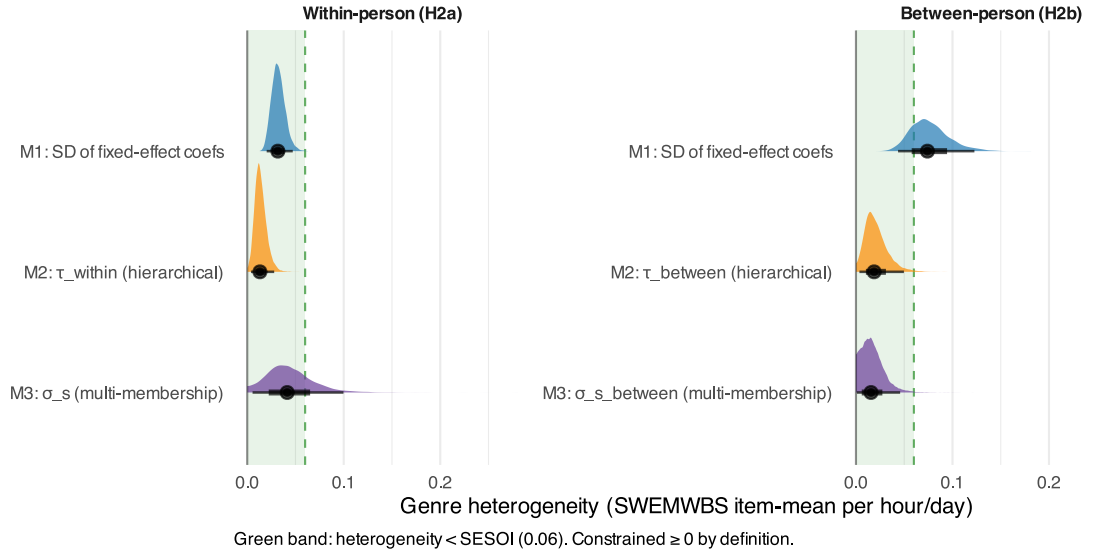


Figure 4. *Genre Heterogeneity Across H2 Model Specifications*. Left: within-person (H2a); right: between-person (H2b). Green band: SESOI region (0 to 0.06). A posterior falling entirely inside the band indicates practically negligible genre heterogeneity.

smallest effect of interest, though the wide interval means heterogeneity of up to roughly 0.1 points per hour/day cannot be ruled out. For the between-person dimension (H2b), $\sigma_{s, \text{between}} = 0.017$ (95% CrI: [0.001, 0.046]), with only 0.5% of the posterior exceeding the SESOI, meaning that between-person genre heterogeneity is practically negligible.

Genre variation in within-person slopes (H2a) is practically negligible under the hierarchical prior model (M2): $\tau_{\text{within}} = 0.014$ (95% CrI: [0.004, 0.028]), with 100% of the posterior inside the ROPE (± 0.06). The results are the same for the between-person dimension (H2b), $\tau_{\text{between}} = 0.021$ (95% CrI: [0.004, 0.05]), with 99.2% of the posterior inside the ROPE, showing that individuals who habitually favour certain genres do not meaningfully differ in average wellbeing from those who favour others.

Finally, the pre-registered fixed-effects specification (M1) provides the simplest version of the test. We computed the standard deviation of the 14 genre-specific within-person coefficients at each posterior draw: every iteration of the Markov chain Monte Carlo sampler yields one full set of the 14 coefficients, and therefore one SD, so the spread across genres is itself estimated with uncertainty attached. The posterior mean of this heterogeneity SD was 0.032 (95% CrI: [0.02, 0.047]), an interval that lies entirely below the SESOI of 0.06. For between-person effects the heterogeneity SD was 0.076 (95% CrI: [0.044, 0.123]), the only statistic among the three specifications to exceed the SESOI; as discussed below, this raw SD includes sampling error and should be read as an upper bound.

The H2 heterogeneity estimates converge on a consistent qualitative picture: genre-to-genre variation in per-hour wellbeing slopes is practically negligible and at most small. On the within-person dimension, M2's τ (0.014) and M1's raw SD (0.032) both fall entirely below 0.06, while M3's σ_s (0.044) straddles the boundary with about a fifth of

its posterior above it. On the between-person dimension, the two pooled estimates (M2, M3) fall below the SESOI while M1's raw SD exceeds it. M1's SD is the empirical standard deviation of 14 independently estimated coefficients and therefore includes estimation error, especially from rare genres with wide posteriors; M2's τ and M3's σ_s are model-estimated pooling parameters that separate signal from noise. Overall, the pooled specifications place genre heterogeneity below the pre-registered equivalence threshold on both dimensions, and no specification provides positive evidence of practically meaningful genre differences.

The heterogeneity estimates and convergence diagnostics for all three specifications are tabulated in Appendix E (Table A0.1). Full posterior summaries for all four models are provided in the supplementary materials.

Per-Genre Estimates

Figure 5 shows the per-genre exposure coefficients from the three H2 specifications, with the within-person and between-person dimensions in side-by-side panels; each genre row shows M1 (circle), M2 (square), and M3 (diamond). We describe each model's estimates in turn before comparing them.

Under the fixed-effects specification (M1), the 14 within-person point estimates cluster near zero, and 8 of the 14 credible intervals fall entirely inside the ROPE; the remainder straddle a boundary rather than exclude it. The cost of independent estimation is visible in the interval widths: the widest within-person interval (Point-and-Click) is roughly 5 times wider than the narrowest, too wide to support any genre-specific conclusion for rarely played genres. Between persons the intervals are wider still, with 2 of the 14 entirely inside the ROPE.

The hierarchical prior model (M2) pulls these estimates towards their common centre: 14 of the 14 within-

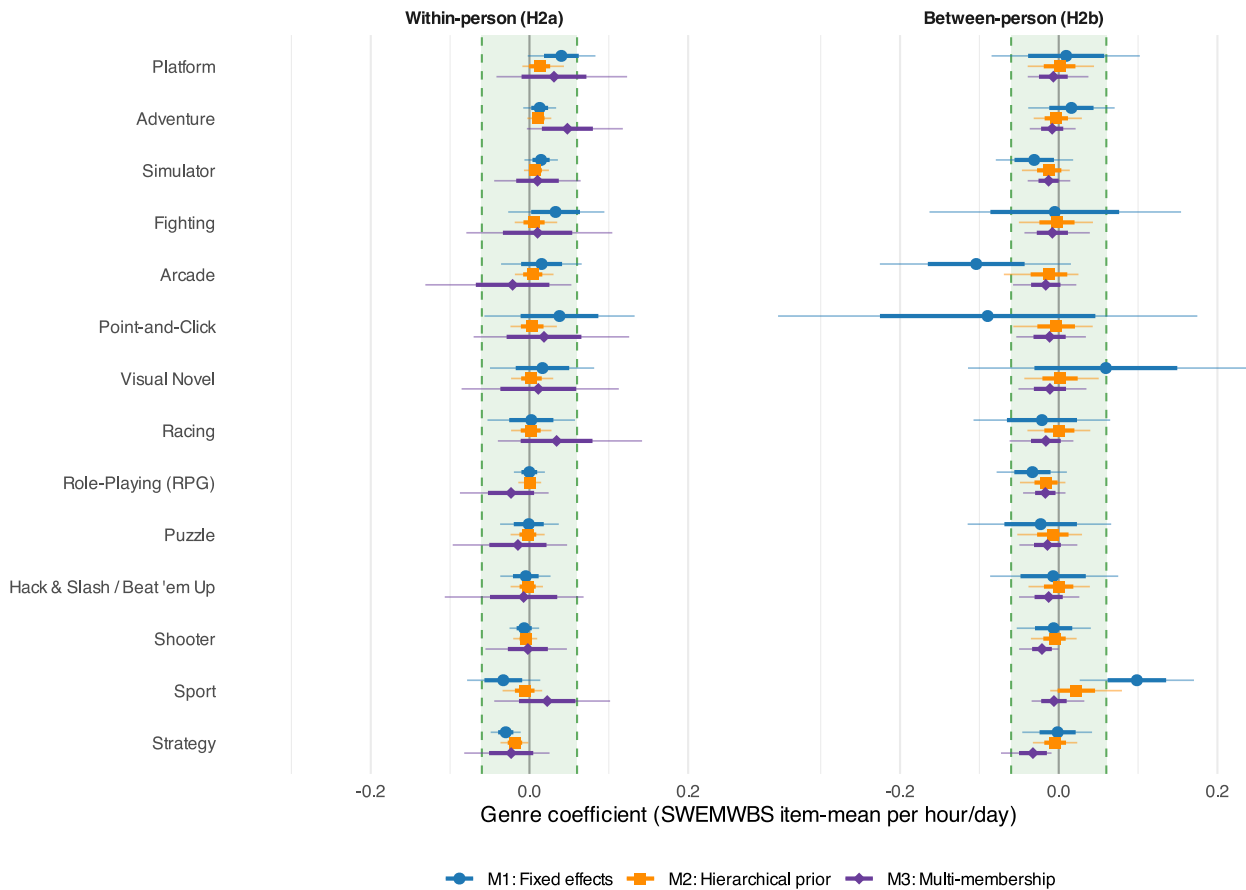


Figure 5. *Per-Genre Exposure Coefficients Across Three Model Specifications*. Left: within-person (H2a); right: between-person (H2b). Each genre row shows M1 (circle, no pooling), M2 (square, hierarchical prior), and M3 (diamond, multi-membership). Genres ordered by the M2 within-person estimate. Green band: pre-registered ROPE (± 0.06).

person genre coefficients have 95% CrIs that fall entirely within the pre-registered ROPE band (± 0.06). The between-person coefficients are wider but show a similar pattern, with Sport as the genre whose point estimate sits closest to the upper ROPE boundary.

The multi-membership model (M3) estimates each genre as a deviation from the overall playtime slope; with the overall fixed effect added back, its within-person estimates span -0.023 to 0.048 and its between-person estimates -0.033 to -0.006 , with 1 and 12 of the 14 intervals respectively falling entirely inside the ROPE.

The three model specifications thus converge on a consistent per-genre pattern for both within- and between-person effects. The primary visible difference is that M2 and M3 produce narrower credible intervals for rare genres (Visual Novel, Fighting, Point-and-Click) relative to M1's wider independent estimates, reflecting the partial pooling that each achieves through different mechanisms. For common genres, the three specifications produce near-identical point estimates and interval widths.

Taken together, the H2 results do not support the pre-registered prediction that genres differ meaningfully in their association with wellbeing. The pooled heterogeneity estimates fall below the smallest effect size of interest on

both the within- and between-person dimensions, genre terms do not improve out-of-sample prediction (Appendix D), and the wide credible intervals for rare genres preclude confident inference about any specific genre–wellbeing pair.

Robustness

Three sensitivity analyses probe the conclusions' dependence on analytic choices; all three leave every conclusion intact. Adjusting for age, gender, and ethnicity shifts the H1 coefficients by less than 0.01 points and changes no ROPE verdict (Appendix D, Table A0.1). Halving or doubling the heterogeneity hyperprior scales leaves the H2 parameters essentially unchanged (Appendix D, Table A0.2). Refitting on multiply imputed data ($m = 100$) reproduces the H1 equivalence under the pre-registered 90% CI rule (equivalence tests are two one-sided tests at $\alpha = .05$, which correspond to a 90% rather than a 95% interval; Lakens (2017)) and rejects genre-coefficient equality in neither pooled dimension (Appendix D, Table A0.4). Full specifications, results, and diagnostics for all three analyses, together with the predictive model comparison and a description of the pre-registered frequentist analysis supplement, are reported in Appendix D.

Exploratory Results: The Daily Timescale

Participants in the analysed daily sample contribute a median of 20 diary days, and participation declines from 1,185 responses on day 1 to 411 on day 30. Windows have a median length of 13.7 hours and close at the survey completion timestamp, so 60.8% of person-days record zero console/PC play in the window. Both positive controls pass in this subsample: self-reported past-24-hour play agrees with telemetry in a matched 24-hour window on 70.7% of responses ($\phi = 0.39$), and men play more Shooter than women ($d = 0.27$).

Figure 6 displays the posteriors against the ROPE. At the daily level, the within-person association between same-day playtime and life satisfaction was positive, at $\beta = 0.18$ points per additional hour in the standalone total-playtime model (M0; 95% CrI [0, 0.35]), with the lower credible bound at zero and the entire posterior inside the ROPE (100%); the multi-membership model's fixed effect agrees (M3: $\beta = 0.14$, 95% CrI [-0.19, 0.42]). The interval widens past zero under two of the three robustness refits (Appendix D). Person-to-person variation in this slope was practically negligible: the random-slope SD was 0.64 (95% CrI [0.18, 0.97]), with 98.3% of the posterior below the SESOI. Between persons, with habitual play measured in full-day hours, participants who play more than others reported slightly lower daily life satisfaction: M0 gives $\beta = -0.43$ points per habitual daily hour (95% CrI [-1.11, 0.27]), a negative lean whose posterior sits 95% inside the ROPE; M3 gives $\beta = -0.46$ (95% CrI [-1.18, 0.27]); and the all-responses refit, which scores unobserved play as zero, gives $\beta = -0.52$ (95% CrI [-1.18, 0.15]) (Table A0.2).

Genre heterogeneity at the daily level followed the same pattern. Within persons, all three specifications place genre heterogeneity inside the ROPE: the unpooled M1 gives an SD across the 14 genre coefficients of 0.61 (95% CrI [0.39, 0.85]; computed at each posterior draw and noise-inflated, as at the biweekly level, yet with 99.8% of its posterior below the SESOI), the hierarchical prior gives $\tau_{\text{within}} = 0.15$ (95% CrI [0.01, 0.43]; 100% below the SESOI), and the multi-membership model gives $\sigma_{s,\text{within}} = 0.32$ (95% CrI [0.01, 0.98]; 97.7% below the SESOI). The between-person results are inconclusive rather than negligible: $\tau_{\text{between}} = 0.84$ (95% CrI [0.09, 1.86]) straddles the SESOI, with 67.3% of its posterior below it; $\sigma_{s,\text{between}} = 0.19$ (95% CrI [0.01, 0.68]) keeps 99.5% of its posterior below it; and M1's between-person raw SD (2.37, 95% CrI [1.48, 3.5]) sits far above it, with the usual estimation-error inflation.

Discussion

When previous work has reported that time spent gaming carries no meaningful association with wellbeing, the objection, from skeptics, to parents, to clinicians, and players alike, is that surely it must depend on the type of game one plays. After all, a relaxing puzzle game and a competitive shooter should not plausibly have the same effect on the person playing them. Video game heterogeneity is the standing explanation for why aggregate playtime nulls should not be taken at face value, since a null total association can be produced by heterogeneous true effects that offset one another in aggregate (Ballou, Hakman, et al., 2025; Yarkoni, 2019). We found this argument persuasive

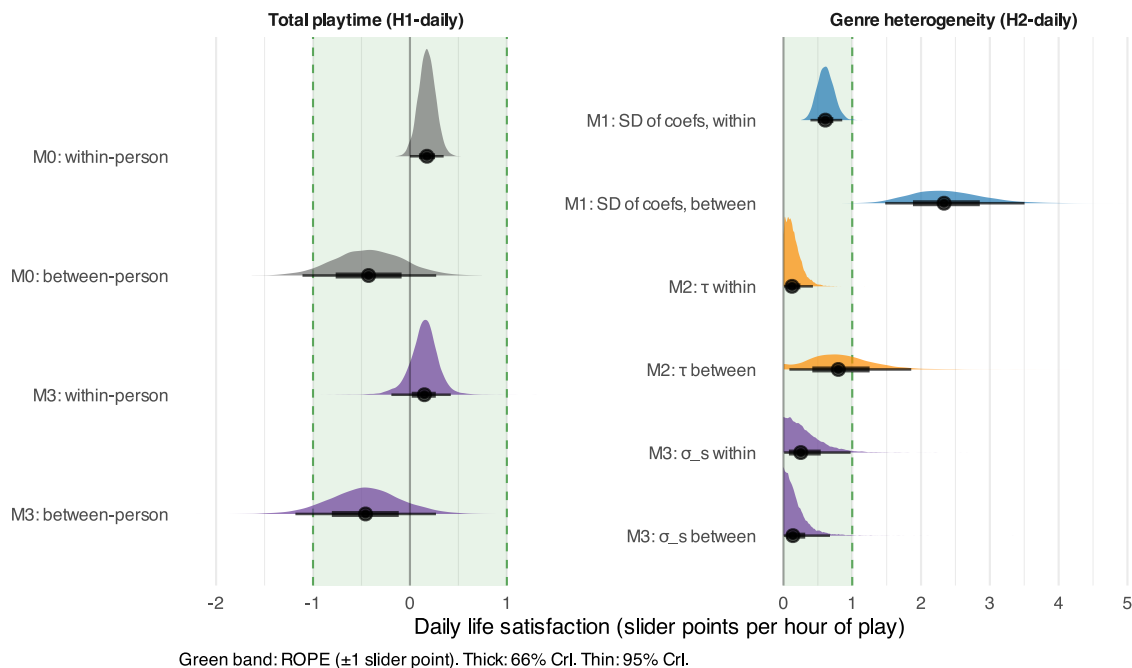


Figure 6. *Exploratory Daily-Timescale Posteriors (Life Satisfaction)*. Left: total-playtime effects from the standalone H1 model (M0) and the multi-membership model's fixed effects (M3). Right: genre heterogeneity from all three specifications: M1 (SD across the 14 unpooled coefficients, computed at each posterior draw), M2 (hierarchical-prior τ), and M3 (multi-membership σ_s). Green band: ROPE (± 1 slider point per hour; 0 to 1 for the non-negative heterogeneity parameters).

enough to register it as our hypotheses, expecting overall playtime to have no effect on wellbeing, but for genres to differ. This study was built to answer that question.

The question has stood open for at least three decades, since Griffiths (1993) reviewed the evidence on ‘computer games’, not for lack of interest but for lack of measurement. Many answers have been proposed (ref) and refuted (ref), it has even impacted policy decisions and public opinion (refs needed). However, all prior genre research carried at least one of three limitations, and usually more than one at once. First, genre assignment is often subjective, roughly half of the genre studies employ a bespoke classification, one in five never state the basis of its categories, and labels are typically self-assigned by participants or researchers, meaning the same games are habitually classified into different genres across studies (Starosta et al., 2024). Second, playtime has often been self-reported rather than logged, and self-reported and logged playtime correlate positively but far from perfectly (our own positive control puts the biweekly correlation at $r = 0.469$) (ref ellis). Third, Research designs often cover a single game or a handful of titles (Brühlmann et al., 2020; Johannes et al., 2021; Larrieu et al., 2023), leaving genre contrasts to be assembled across different studies that classified and measured both play and genre incommensurably, instead of one study with a single ground truth. This study clears all three obstacles at once. Measuring objective session-level telemetry across Nintendo, Xbox, and Steam; utilising one crowd-sourced genre taxonomy applied uniformly to 94.1% of in-study play, estimating 14 genres jointly in one sample with models capable of analysing games carrying multiple genre labels. Making it, to our knowledge, the first study positioned to answer the questions directly.

The playtime-wellbeing association is practically equivalent to zero

Does playtime in general have an effect on wellbeing? H1 predicted not, and both components of the hypothesis were supported (see Figure 3 and Table A0.1). Within persons, when participants played more than their own average in a given two-week wave, their wellbeing barely changed, consistent with previous digital trace studies that have repeatedly found aggregate playtime to be a poor predictor of wellbeing fluctuations (Ballou, Sewall, et al., 2024; Ballou, Vuorre, et al., 2025; Larrieu et al., 2023). Between persons, practical equivalence was likewise established.

Inverting these estimates makes their practical magnitude concrete. The determined SESOI derives from the smallest wellbeing change a person can themselves notice (Ballou, Sewall, et al., 2024; Maheswaran et al., 2012), divided by the roughly 5 hours of daily leisure realistically available for reallocation (Sturm & Cohen, 2019). Against that anchor, the estimates are not merely small; the doses they imply are unreachable. For wellbeing to shift by one minimally important difference, the within-person point estimate requires roughly 98 additional hours of play per day, several times more hours than a day contains, and even the credible bound most favourable to an effect requires 22 extra daily hours, more extreme than the 10

additional daily hours that Vuorre & Przybylski (2022) derived from game-level telemetry. These doses are so small the even the sample’s variation cannot contain them: of the 1,500 participants observed across adjacent waves, only 121 ever changed their average daily play by 4 or more hours from one wave to the next, and the largest swing recorded anywhere in the sample was 9.3 hours per day. And if a player reallocated the entire 5-hour daily leisure budget into or out of gaming, the implied shift of about 0.015 points would still sit far below the roughly 0.2 points of measurement error that surround an individual’s score on the parent WEMWBS [1 SEM; Maheswaran et al. (2012)]. An association that no achievable dose can reach and that its own instrument cannot accurately measure is, for every practical purpose, indistinguishable from no association at all.

Genres do not matter for the playtime-wellbeing association

We expected genres to differ in how playtime relates to fluctuations in general mental wellbeing, however the findings disconfirmed our registered hypothesis. The point estimates of genre-to-genre variation in the within-person per-hour association fell at or below the smallest effect we deemed practically meaningful in all three model specifications; the 14 within-person genre estimates span only -0.018 to $+0.013$ under the hierarchical prior, with no estimate more than a third of the SESOI from zero, and the most-played genres by total playtime (Role Playing Game (RPG), Adventure, Strategy, Shooter; see Figure 1) sit inside this same narrow band. The complete-case joint test does detect a departure from exact within-person coefficient equality, but of a magnitude well below the SESOI (Appendix D). Meaning that, genres proved statistically distinguishable but practically interchangeable. One notable exception is M3’s σ_s , which retains roughly a fifth of its posterior mass above the SESOI, so within-person heterogeneity slightly beyond the threshold cannot be fully excluded. Thus, the effect of separate genres canceling out a general playtime effect is below the threshold of noticeability, and any defence of playtime nulls that appeals to the heterogeneous nature of video games must look to partitions finer than genre.

Who Plays, Not What Playing Does

The contrast between light and heavy players is a different quantity from either hypothesis test. Participants who habitually play more reported slightly lower average wellbeing, however the same participants’ wellbeing did not fall in waves when they played more than usual. This effect is small in practical terms, two otherwise-similar people would need to differ by between roughly 9 hours of habitual daily play (at the credible bound most favourable to an effect) and 18 hours (at the point estimate), in either case more than the roughly 5 hours of daily leisure available, before their expected wellbeing differed noticeably. Its interpretation matters more than its size. That within-null / between-negative pattern is a potential selection effect, people with lower baseline wellbeing spending more time in games rather than games lowering wellbeing, though a

cumulative effect accruing over years would look the same at these timescales (cf. Vuorre & Przybylski, 2022); it recurs across observational media research (Ballou, Vuorre, et al., 2025; Kievit et al., 2013; Robinson, 1950; Rohrer & Murayama, 2023). The genre-level results carry the same fingering. Descriptively, the largest positive between-person point estimate belongs to Sport (0.022, 95% CrI [-0.011, 0.08] in M2), yet its within-person counterpart is null (-0.006, 95% CrI [-0.034, 0.016]): people who favour sport games may, for example, be more physically or socially active, rather than playing the sport genre itself improving wellbeing, consistent with exploratory work concluding that players select the games that align with their personal sources of enjoyment (Griffith & Sharpe, 2024). The best available causal evidence points in the opposite direction, console-lottery estimates from Egami et al. (2024) are positive, 0.1 to 0.6 standard deviations, on data whose own observational analysis correlates negatively. Our within-person null estimates, which remove stable person-level confounding, demonstrate that public claims about gamers' mental health are, in effect, reports about who plays, not about what playing does.

Individual Susceptibility Is Bounded at These Timescales

Still, a null average could conceal a minority for whom play genuinely matters (cf. Karhulahti et al., 2025). The random slope in M0 addresses this directly by modelling every participant's own within-person association between playtime and wellbeing. The standard deviation of these person-specific slopes describes how much people differ from the average. That standard deviation was 0.035 (95% CrI: [0.005, 0.062]), with only 3.8% of its posterior above the SESOI. Individuals in the far tails of that distribution may still exceed the SESOI, but we would argue that they are the exception the spread allows rather than a subgroup the data reveal. Two boundaries on this claim are required. These are bounds on general wellbeing at biweekly resolution and on life satisfaction at daily resolution, among adults screened into the study at ages 18 to 40; they say nothing about disordered play, where genre-linked structural characteristics remain plausibly relevant (André et al., 2024; Rehbein et al., 2021), and nothing about children. Future work should address these questions.

Shortening the Window Does Not Recover an Effect

Lastly, a final possible objection is temporal, namely that the wellbeing effects of play might surface and dissipate faster than a biweekly retrospective survey can register. To test this claim, the exploratory daily analysis shortens the window from fourteen days to one and reproduces both conclusions. Within persons, every estimate lies inside the equivalence region under every specification, and the same-day point estimate is positive, roughly double its biweekly counterpart in within-person standard-deviation units (0.014 versus 0.009 s_w per hour). The direction is consistent with the session-level affective uplift reported among *PowerWash Simulator* players (Vuorre et al., 2024) and with the positive causal estimates of Egami et al. (2024), though its resolution is fragile across the robust-

ness refits and should be read as suggestive rather than resolved. Its practical magnitude is negligible either way: shifting daily life satisfaction by one minimally important difference (5 points) would require 28 additional hours of same-day play, more hours than a day contains, and within-person genre heterogeneity at the daily level is likewise bounded (its between-person counterpart is inconclusive rather than negligible). Therefore, moving the measurement closer in time, on its own, does not conjure a meaningful effect.

Are all games the same?

The question this study set out to answer rests on an intuition, that games differ psychologically, and although genre labels are not the way to find that difference, the intuition survives the answer. What fails is its operationalisation. Game scholars have argued for decades that genre labels are conventions of marketing and library organisation rather than coherent descriptions of play: they conflate gameplay style, purpose, and mood in a single word (Lee et al., 2014), fail elementary classificatory tests (Clarke et al., 2017), and drift as the medium evolves (Cohen, 1986; Vargas-Iglesias, 2020). The public, policymakers, and psychological researchers, imported the labels anyway, and the present results show what they are worth as a predictor of wellbeing. This study has provided the fairest test possible to find genre-specific effect, and found that genre carries no meaningful effect. The pathways plausibly connecting play to wellbeing seem to operate at levels that genre labels cannot index (Ballou, Hakman, et al., 2025): need satisfaction depends on autonomy-supportive design features (Ballou et al., 2023; Peng et al., 2012), social capital depends on co-play relationships (Depping et al., 2018; Vella et al., 2015), and affect regulation depends on the match between gameplay and current emotional state (Kosa et al., 2020). Dedicated taxonomies of structural game features now exist for characterising these directly (Saini & Hodgins, 2023). And within a single game, behavioural variation in how people play is associated with meaningful wellbeing differences while genre is held constant (Hakman et al., 2026). If wellbeing differs among people playing the same game, a label that treats those people as identical is potentially measuring the wrong thing. The person who objects that a puzzle game and a shooter cannot do the same thing is right that the games differ; but the effect of playing them on wellbeing does not.

Policy implications

For public guidance, the implication is uncomfortable because hours are the quantity institutions know how to handle, time is the property of play most readily measured, displayed, and limited, and it is where wellbeing guidance therefore concentrates (ref the screentime bans). Our estimates predict what such instruments can deliver on the margin they operate on, hours among people already playing, if the observed association were causal, shifting daily play by a full hour, in either direction and sustained for weeks, would move wellbeing by an unnoticeable and irrelevant amount. Steering adults between genres is sim-

ilarly unpromising, the heterogeneity estimates describe freely chosen play, and reassigning players to genres they would not choose is a different intervention, one these data cannot speak to. If the aim is wellbeing, playtime is the wrong lever. Read in the other direction the same results are actually reassuring. Within the range of ordinary adult play we observed, no specification provides positive evidence of a practically meaningful wellbeing cost for any genre, though intervals for rarely played genres remain wide, popular worries about particular genres receive no support at these margins.

Methodological Contribution: A Model for Overlapping Categories

Methodologically, the multi-membership model transfers beyond games. To our knowledge this is the first application in media psychology of the model class, developed originally for educational data where students attend multiple schools (Cafri et al., 2015; Fielding & Goldstein, 2006). It treats a game's genres the way the data present them, as overlapping memberships with composition weights, rather than forcing single-category assignment or independent fixed effects. Partial pooling through σ_s stabilises sparse categories, and the specification was validated by parameter recovery before touching real data. Any media effects question with overlapping content labels has the same structure (e.g., streaming titles with multiple genre tags, news diets spanning outlets with overlapping orientations, social media feeds spanning topic categories) and can be answered with the same model. The code was open-sourced and made freely available to support future applications of this method.

Limitations

Several limitations constrain these conclusions and point at the work that should come next. First, the design is observational, the within-person decomposition removes stable confounds but not time-varying ones (life events, seasons), and our causal statements are correspondingly guarded (Rohrer & Murayama, 2023). Second, the SWEMWBS is a global instrument assessed over a two-week recall window, gaming competes with sleep, work, and social life for within-person wellbeing variance, and domain-specific outcomes (affect, sleep, connectedness) could show associations a global measure averages away. The daily analysis takes the first step down this temporal hierarchy and still finds nothing of practical size, which only sharpens rather than settles the point. Future work should pair session-level telemetry with momentary assessment, and use time-use data to partial gaming from competing activities. Third, coverage is imperfect, the achieved platform-coverage criterion admitted participants with as little as half of their play observed, mobile play carries no genre metadata, and in-study minutes are dominated by Steam (84.1%, versus 12.8% Xbox and 3.1% Nintendo), so the genre findings are predominantly findings about PC play by UK and US adults. Fourth, rare genres were estimated imprecisely, their effects remain untested rather than refuted, a data sparsity problem that larger samples or finer temporal resolution could solve.

Data and Code Availability

The Open Play dataset (v1.1.0) is openly available on Zenodo at <https://zenodo.org/records/18430947>, released under a CC0 licence with a supplemental no-reidentification clause (Ballou, Földes, et al., 2025). All analysis code is available at <https://github.com/Thomhak/open-play-genres>. Pre-registration materials for the Stage 1 programmatic registered report are available at <https://osf.io/mvngt/>.

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Conflicts of Interest

The authors declare no competing financial, intellectual, or institutional interests relevant to this research. The disclosures below are provided in the interest of full transparency.

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Data-sharing partners had no role in the design, analysis, or publication of results.

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A1 Appendix A: Genre Mapping

Genre labels from all three platforms resolve to the study taxonomy in two steps: Xbox store labels are first mapped onto IGDB genres (per the pre-registration’s Appendix B), and raw IGDB labels (23 distinct labels observed in the study metadata) are then collapsed into 14 analysis-level categories. Table A1.1 presents both steps in one place: each row is an implemented study genre, with the raw IGDB labels it subsumes, the Xbox store labels routed to it, and, where the pre-registered mapping (which would yield 17 distinct genres on the observed labels) treated a label differently, the pre-registered assignment. The implemented mapping diverges from the pre-registered one in both directions: it preserves granularity for sub-genres that prior literature treats as distinct (Hack-and-slash/Beat-‘em-up, Point-and-click, Visual Novel), collapses more aggressively elsewhere (MOBA into Strategy, Card & Board Game into Puzzle, Pinball and Music into Arcade), and excludes Indie as a non-substantive label (see Appendix B, Deviation D2). Two details sit outside the table: the Xbox label Family + Kids has no IGDB equivalent and is resolved by the title’s secondary genre, and IGDB includes no top-level “Action” category, so what is colloquially called “action” is distributed across Shooter, Hack-and-slash, Fighting, and Platform.

Study genre	IGDB label(s) subsumed	Xbox store label(s)	Pre-registered assignment (where different)
Adventure	Adventure	Action + Adventure	Also subsumes Point-and-click and Visual Novel
Arcade	Arcade, Pinball, Music	Classics, Music	Pinball and Music retained as separate genres
Fighting	Fighting	Fighting	
Hack-and-slash/Beat-‘em-up	Hack and slash/Beat ‘em up		Collapsed into Action
Platform	Platform	Platformer	
Point-and-click	Point-and-click		Collapsed into Adventure
Puzzle	Puzzle, Quiz/Trivia, Card & Board Game	Puzzle + Trivia, Card + Board, Casino	Card & Board Game retained as a separate genre
Racing	Racing	Racing + Flying	
Role-playing (RPG)	Role-playing (RPG)	Role Playing	
Shooter	Shooter	Shooter	
Simulator	Simulator	Simulation	
Sport	Sport	Sports	
Strategy	Strategy, Turn-based strategy (TBS), Real time strategy (RTS), Tactical, MOBA	Strategy, Multi-Player Online Battle Arena	MOBA retained as a separate genre
Visual Novel	Visual Novel		Collapsed into Adventure
(excluded)	Indie		Retained as Indie

Table A1.1. Mapping from platform genre labels to the 14 study genres. Xbox store labels map onto IGDB genres, which are then collapsed into the study taxonomy. The final column gives the pre-registered assignment where it differs; blank cells indicate identical treatment.

A2 Appendix B: Deviations from Pre-Registration

This appendix documents all known deviations from the Stage 1 pre-registration (Ballou, Földes, et al., 2024) in a single numbered scheme (D1–D9). Every deviation is presented in the structured disclosure format of Willroth and Atherton (2024), reporting the type of deviation, the original pre-registered wording, a description of what was done instead, the reason, the timing relative to data access and results, and the potential impact on interpretation. Related decisions are documented together rather than split across entries: prior specification is part of the inference-framework deviation (D4), the H2 model-structure decisions form a single deviation (D5), and the three changes that arose upstream during data collection (outside the authors’ control and reported in full by Ballou, Földes, et al. (2025)) are consolidated as D9. One unregistered correction, which implements the pre-registered plan more faithfully than the upstream data pipeline did, is documented last in the same format (U1).

#	Topic	Type	Timing
D1	Genre assignment (full attribution)	Data preparation	After data access, before results known
D2	Genre taxonomy (sub-genre collapsing)	Data preparation	After data access, before results known
D3	Telemetry–survey alignment	Variables	After data access, before results known
D4	Inference framework and priors	Inferential criteria	After data access, before results known
D5	H2 model specifications	Analytic approach	Partly after H2 results known
D6	H1 platform scope	Variables	After data access, before results known
D7	Outcome measure and scoring	Variables	Instrument: during data collection; scoring: after data access
D8	Missing data handling	Analytic approach	Protocol upgraded after complete-case results known
D9	Upstream data-collection changes	Data collection procedures	During data collection, before data access
U1	Xbox genre mapping correction	Data preparation (correction)	After data access, before results known

A2.1 Structured disclosures

A2.1.1 D1: Genre assignment (full attribution)

Field	Content
Type	Data preparation
Original wording	Games are classified by their primary (first) IGDB genre. The pre-registration does not specify how to handle multi-genre games beyond this.
Deviation description	Each game’s full playtime is attributed to every one of its IGDB genre labels (all labels retained; up to 16 raw labels per game observed in the metadata). A game with 2 hours of play and two genre labels (e.g., Adventure + RPG) contributes 2 hours to each genre. Genre hours therefore represent total exposure to games carrying that label and are not mutually exclusive: summing across genres exceeds total playtime. Total playtime for H1 is computed separately from raw sessions without genre duplication. An earlier draft capped attribution at 3 genres per game; the cap was removed because

Field	Content
	25% of games with any genre label carry more than 3 raw IGDB labels and truncation discards classification information.
Reason	Many games genuinely span multiple genres; restricting to the primary genre discards meaningful classification information and introduces arbitrary dependence on IGDB's label ordering. Full attribution reflects actual exposure to each genre.
Timing	After data access, before results of the pre-registered approach were known
Reader impact	Full attribution means genre-level playtime variables cannot be summed to recover total playtime, and each genre coefficient estimates the association with exposure to games carrying that label rather than the effect of the genre in isolation. This is appropriate because a player of an Adventure-RPG is fully exposed to both Adventure and RPG design elements. Total playtime is tracked independently to avoid conflating genre exposure with total gaming time. The pre-registered primary-genre analysis is reported in the supplement.

A2.1.2 D2: Genre taxonomy (sub-genre collapsing)

Field	Content
Type	Data preparation
Original wording	The pre-registration collapses IGDB sub-genres as follows: Hack and slash/Beat 'em up → Action; Point-and-click and Visual Novel → Adventure; Turn-based strategy, Real-time strategy, and Tactical → Strategy; Quiz/Trivia → Puzzle.
Deviation description	Hack-and-slash/Beat-'em-up, Point-and-click, and Visual Novel are retained as distinct categories; MOBA is folded into Strategy; Card & Board Game joins Quiz/Trivia under Puzzle; Pinball and Music are folded into Arcade; Indie is excluded as a non-substantive label. The implemented taxonomy comprises 14 analysis-level categories (full mapping in Appendix A of this manuscript). The metadata contain 23 distinct raw IGDB labels; the pre-registration's body text mentions 19 IGDB genres while its own Appendix A lists all 23, and we report the count as observed in the data.
Reason	The pre-registered collapsing conflated meaningfully different genres; e.g., Visual Novels and Beat-'em-ups would both disappear into broad categories despite prior literature treating them as psychologically distinct. Retaining them preserves granularity where theory suggests it matters.
Timing	After data access, before results of the pre-registered approach were known
Reader impact	Genre-level coefficients refer to different category boundaries than the pre-registration implies, so per-genre estimates are not directly comparable to a hypothetical analysis under the original taxonomy. The retained sub-genres are rarely played, producing wide credible intervals for those categories; the pooled heterogeneity parameters (M2, M3) are designed to remain stable under this sparsity.

A2.1.3 D3: Telemetry–survey alignment

Field	Content
Type	Variables
Original wording	Not specified; the pre-registration implicitly assumes wave windows counted forward from the baseline day (days 0–83).
Deviation description	Playtime windows are aligned retrospectively to the survey: each wave's playtime covers the 14 days ending the day before the biweekly survey, so wave 1 covers days –14 to –1 relative to baseline. Pre-baseline telemetry is available for all three platforms because participants linked their accounts at intake.

Field	Content
Reason	SWEMWBS asks participants to reflect on “the past 2 weeks.” The wave 1 survey is completed on the baseline day, so its retrospective window predates the study start; forward-counted windows would misalign every wave’s playtime with the period participants were actually reporting on.
Timing	After data access, before results of the pre-registered approach were known
Reader impact	Improves measurement validity by matching exposure and outcome windows. Estimates are not directly comparable to an analysis using forward-counted windows, and wave 1 relies on telemetry recorded before the study formally began (over 85% of participants per platform had observable telemetry covering the full wave 1 period).

A2.1.4 D4: Inference framework and priors (frequentist → Bayesian)

Field	Content
Type	Inferential criteria
Original wording	H1 will be tested via equivalence test with SESOI = ± 0.06 SWEMWBS points per 1-hour change in daily playtime, using 90% CIs (TOST procedure). H2 will be tested via a joint Wald test via <code>car::linearHypothesis()</code> . The pre-registration specifies no priors (all models frequentist).
Deviation description	Both H1 and H2 are estimated using Bayesian multilevel models [brms; Bürkner (2017)] rather than frequentist <code>lmer</code> . H1 is evaluated via ROPE analysis with bounds identical to the pre-registered SESOI $[-0.06, +0.06]$; if >95% of the posterior falls within the ROPE, practical equivalence is established (Kruschke, 2018). H2 is evaluated through posterior genre-heterogeneity parameters (σ_s in M3, τ in M2, the SD of coefficients in M1), with magnitude judged against the SESOI. Because the pre-registration specifies no priors, these were chosen as weakly informative on the 1–5 item-mean outcome scale: Student- $t(3, 0, 0.5)$ on fixed-effect coefficients, Student- $t(3, 3, 2)$ on the intercept, Half-Normal(0, 0.1) on M2’s τ hyperparameters, Half-Normal(0, 0.3) on M3’s σ_s (looser to accommodate the more challenging multi-membership posterior geometry), and brms defaults on remaining variance components. The pre-registered frequentist models (TOST and joint Wald tests) are reported in full in the pre-registered analysis supplement.
Reason	A Bayesian framework provides richer equivalence testing via ROPE and continuous uncertainty quantification; the priors regularise estimation without strongly constraining the posterior.
Timing	After data access, before results of the pre-registered approach were known
Reader impact	The SESOI is unchanged and the ROPE procedure provides a continuous measure of practical equivalence rather than a binary significance decision. Risk of bias is low: the equivalence region is identical and the deviation was decided before results were known. More diffuse priors could permit larger heterogeneity estimates; prior sensitivity could be assessed by varying the prior scales.

A2.1.5 D5: H2 model specifications

Field	Content
Type	Analytic approach
Original wording	H2 uses genre-specific fixed-effect coefficients with a random intercept and uncorrelated random slopes for all within-person genre predictors (zero-correlation syntax), tested via a joint Wald test. The pre-registration anticipates identifiability issues and specifies dropping random slopes if the model is singular.

Field	Content
Deviation description	Three related decisions. (a) The pre-registered random slopes (14 genres $\times$ 2 dimensions = 28 slope parameters) render the model near-singular; the random-intercept-only contingency was applied, as pre-registered. (b) Two additional specifications were added: M2 replaces the independent priors on genre coefficients with a learned hierarchical prior (τ), and M3 is a multi-membership random-effects model in which genre modulates the total playtime slope via composition weights, adopted as the primary H2 analysis. (c) For M3, each wave's membership weights are that wave's genre shares of total genre-attributed playtime; waves with zero playtime receive uniform weights ($1/K$), a technical requirement of <code>brms::mm()</code> .
Reason	With 28 independently estimated genre coefficients, the pre-registered specification is overparameterised and rare-genre estimates are dominated by sampling noise. The pooled specifications (M2, M3) separate genuine genre heterogeneity from estimation error and answer H2 through a single interpretable parameter.
Timing	The random-slope contingency was applied at model fitting. M2 and M3 were added after the fixed-effects (M1) results were known.
Reader impact	The timing of adding M2 and M3 (after seeing M1 results) increases risk of bias for the choice of primary analysis; readers who prefer the pre-registered specification can rely on the M1 results, reported in full, which lead to the same qualitative conclusion. The M3 specification was validated with a parameter-recovery simulation before being fit to real data. Uniform weights for zero-playtime waves have minimal substantive impact, as those observations contribute little information to genre-specific estimates.

A2.1.6 D6: H1 platform scope (mobile excluded)

Field	Content
Type	Variables
Original wording	H1 will test the association between total cross-platform playtime and wellbeing, using data from all five platforms (Nintendo, Xbox, Steam, iOS, Android).
Deviation description	H1 total playtime is computed from Nintendo, Xbox, and Steam only, excluding iOS and Android. Mobile platforms provide only daily aggregate minutes without game-level data (D9). Including mobile for H1 but not H2 would mean "total playtime" refers to different quantities across hypotheses.
Reason	Internal consistency between H1 and H2.
Timing	After data access, before results of the pre-registered approach were known
Reader impact	This reduces the total playtime measure by excluding mobile gaming. However, mobile contributed a small minority of total playtime for most participants. The pre-registered analysis supplement retains the all-platform total, allowing a direct comparison of console/PC-only versus all-platform H1 estimates. Risk of bias is low: the deviation was motivated by internal consistency, not by the data or results.

A2.1.7 D7: Outcome measure and scoring

Field	Content
Type	Variables
Original wording	General mental wellbeing will be measured with the WEMWBS (Tennant et al., 2007).
Deviation description	The instrument administered in the Open Play study is the short-form SWEMWBS (7 items, each scored 1–5), an upstream design decision documented by Ballou, Földes, et al. (2025). The score is computed as the item mean (rowMeans of the 7 items, range 1–5), not the summed score (range 7–35). The pre-registered SESOI of 0.06 per hour/day

Field	Content
	was derived by Ballou, Sewall, et al. (2024) on the 1–5 item-mean scale (0.3 minimally important difference ÷ 5 hours of realistic daily leisure), and the pre-registration’s own precision analysis assumes a within-person SD of 0.7, which is only coherent on the 1–5 scale. An earlier version of this analysis used rowSums, which inflated coefficients by a factor of 7 relative to the scale the SESOI was defined on.
Reason	Item-mean scoring places the coefficients, priors, and SESOI on a single consistent scale.
Timing	Instrument: during data collection, before data access (upstream). Scoring: after data access, before results were known.
Reader impact	No change to the underlying data or model specification; only the summary score computation differs from a summed-score reading of the pre-registration. All downstream numbers (coefficients, equivalence tests, ROPE bounds) are on the scale the SESOI was designed for.

A2.1.8 D8: Missing data handling

Field	Content
Type	Analytic approach
Original wording	Missing data: multiple imputation via mice package (predictive mean matching, wide format). Report both complete-case and imputed results.
Deviation description	Outcome missingness has two components. Among submitted surveys, 43 responses (0.6% of 6,723 in the analytic sample) are missing SWEMWBS, all with every item missing. The larger component is wave nonresponse: 4,348 of 11,028 person-waves (39.4%) in the analytic sample were never submitted (response declined from 100% at wave 1 to 40% at wave 6). Instead of the pre-registered single-level wide-format PMM, we imputed on the complete participant × wave grid with hierarchical two-level MICE (21 .pmm, participants as clusters, ±1-wave lag/lead terms, fully observed exposures with explicit person means, intake auxiliaries), the protocol shared with the programmatic RR’s other Stage 2 outputs; the number of imputations follows von Hippel’s (2020) rule from the pilot fraction of missing information. The primary Bayesian models remain fit to observed outcomes (valid under MAR given the modelled covariates and random effects); pooled frequentist counterparts on the imputed data are reported in Appendix D. The imputation grid covers the analytic sample; 110 surveyed participants with no valid SWEMWBS in any wave provide no outcome information to anchor imputation and are outside both the analytic sample and the grid.
Reason	The pre-registered wide-format single-level PMM understates the hierarchical structure, and an initial implementation of it could only address item-level missingness among submitted surveys. The grid protocol addresses the dominant wave-nonresponse component and is consistent with the sibling Stage 2 studies.
Timing	Complete-case-first decision: after data access, before results known. Grid-imputation protocol: adopted after complete-case results were known, by mirroring the sibling studies’ protocol rather than tailoring choices to results.
Reader impact	The imputation model conditions on all analysis-model exposures, so it is congenial with (indeed richer than) the analysis models. Equivalence and joint-test conclusions are unchanged under imputation (see Sensitivity Analyses); the pooled between-person interval widens because person means are genuinely uncertain for participants with few observed waves (high fraction of missing information), which Rubin’s rules propagate. If nonresponse depends on unobserved wellbeing beyond the modelled covariates (MNAR), both complete-case and imputed estimates could be biased in the

Field	Content
	same direction; the agreement between them rules out bias from the modelled missingness mechanism, not MNAR.

A2.1.9 D9: Upstream data-collection changes

Field	Content
Type	Data collection procedures
Original wording	Age range 18–75 (UK) and 18–30 (US); $\geq 75\%$ of gaming time on included platforms; Android telemetry collected as session-level exports via ActivityWatch.
Deviation description	Three changes made by the Open Play team during data collection, before this study's data access: (a) screening admitted ages 18–40 in both regions; (b) the platform coverage threshold was lowered to $\geq 50\%$; (c) Android data were collected as daily aggregate minutes via Digital Wellbeing screenshots rather than session-level exports. All three are reported in full by Ballou, Földes, et al. (2025).
Reason	Recruitment and data-collection constraints in the upstream Open Play study; outside the authors' control.
Timing	During data collection, before data access
Reader impact	The narrower age range limits generalisability to older adults. The lower coverage threshold admits participants with a larger share of unobserved play, adding measurement error to total playtime. The Android granularity change restricts Android to daily totals, contributing to the pre-registered all-platform sensitivity analysis only and excluding it from H2 (see D6).

A2.2 Unregistered correction

A2.2.1 U1: Xbox genre mapping correction

Field	Content
Type	Data preparation (correction, consistent with the pre-registration)
Original wording	Xbox genres are mapped onto the IGDB taxonomy via the mapping table in the pre-registration's Appendix B.
Deviation description	The upstream Open Play preprocessing pipeline (<code>label_genres.R</code>) failed to map approximately 67% of Xbox playtime to IGDB genres due to a string-formatting bug (exact-match against multi-value and inconsistently-formatted genre strings). We applied a corrected mapping in our preprocessing using the <code>xbox_primary_genre</code> and <code>xbox_subgenre</code> metadata fields, recovering genre information for an additional 20.7 million minutes of Xbox playtime. This is a fix, not a deviation: the pre-registration explicitly specified the Xbox-to-IGDB mapping, and the correction implements it as intended.
Reason	Implementing the pre-registered mapping faithfully; the upstream bug would otherwise cause systematic genre-data missingness for the largest platform by volume.
Timing	After data access, before results were known
Reader impact	Xbox genre coverage increased from 33% to 80% of playtime, substantially reducing systematic missingness in the genre-level analyses.

A3 Appendix C: Sample and Genre Descriptive Statistics

A3.1 Participant Demographics

Characteristic	Total	US	UK
N	1741	1041	700
Age (years)	26.9 (5.1)	26.8 (5.1)	26.9 (5.1)
Gender			
Man	1142 (65.6%)	656 (63%)	486 (69.4%)
Woman	492 (28.3%)	310 (29.8%)	182 (26%)
Non-binary/Other	104 (6%)	73 (7%)	31 (4.4%)
Ethnicity			
White	1209 (69.4%)	632 (60.7%)	577 (82.4%)
Asian	141 (8.1%)	89 (8.5%)	52 (7.4%)
Black	117 (6.7%)	102 (9.8%)	15 (2.1%)
Mixed/Multiple	179 (10.3%)	152 (14.6%)	27 (3.9%)
Other	50 (2.9%)	45 (4.3%)	5 (0.7%)
Platform			
Nintendo	448 (25.7%)	262 (25.2%)	186 (26.6%)
Steam	1394 (80.1%)	780 (74.9%)	614 (87.7%)
Xbox	167 (9.6%)	167 (16%)	0 (0.0%)
iOS	93 (5.3%)	66 (6.3%)	27 (3.9%)
Android	73 (4.2%)	50 (4.8%)	23 (3.3%)

Values are M (SD) or N (percent). Platform counts are non-exclusive (participants may have data from multiple platforms).

Table A3.1. Participant characteristics by country.

A3.2 Genre Playtime Descriptives

Genre	n games	Total hours	Mean min/ day	Top 5 titles
Role Playing Game (RPG)	1033	93034	58	Baldur's Gate III [4.3%], Final Fantasy XIV Online [3.2%], Stardew Valley [3%], Elden Ring: Nightreign [2.9%], Warframe [2.6%]
Adventure	1795	91460	57	Stardew Valley [3.1%], Warframe [2.6%], Path of Exile [2.2%], Peak [2.1%], The Elder Scrolls V: Skyrim - Special Edition [1.7%]
Strategy	982	79136	50	Counter-Strike 2 [6.3%], Dead by Daylight [5.2%], Baldur's Gate III [5%], Balatro [3.6%], Stardew Valley [3.6%]
Shooter	702	64104	40	Marvel Rivals [12.1%], Counter-Strike 2 [7.8%], Overwatch 2 [3.8%], Warframe [3.7%], Helldivers 2 [3.3%]
Simulator	1069	61217	38	Stardew Valley [4.6%], Umamusume: Pretty Derby [3.8%], RimWorld [3.8%], Schedule I [3.5%], Football Manager 2024 [3%]
Puzzle	585	18717	12	Balatro [15.2%], The Binding of Isaac: Rebirth [8.1%], Magic: The Gathering Arena [4.7%], Slay the Spire [4.4%], The Legend of Zelda: Tears of the Kingdom [4.3%]
Sport	195	14556	9	Umamusume: Pretty Derby [16.1%], Football Manager 2024 [12.8%], Rocket League [12.7%], C3C63F46-0627-4660-AAF5-974831379998 [4.4%], BA2D9D35-CA26-4220-9D7D-C41E318FA5CE [4.1%]
Hack And Slash Beat Em Up	163	11262	7	Path of Exile [18%], Monster Hunter Wilds [12.8%], Palworld [12.7%], Path of Exile 2 [7.8%], Last Epoch [3.5%]
Platform	311	11438	7	Peak [16.8%], MapleStory [11.7%], Terraria [10.7%], Super Smash Bros. Ultimate [5.3%], Blender Bros. [4.9%]
Arcade	244	10586	7	Umamusume: Pretty Derby [22.1%], The Binding of Isaac: Rebirth [14.3%], Deltarune [7%], Mario Kart 8 Deluxe [4.9%], Spacewar [4.5%]
Racing	147	8107	5	Rocket League [22.9%], F-Zero DSX [10.9%], Blender Bros. [6.9%], Grand Theft Auto V Enhanced [6.4%], Mario Kart 8 Deluxe [6.4%]
Fighting	149	5091	3	Super Smash Bros. Ultimate [12%], WWE 2K25 [7.8%], Brotato [5.9%], Tekken 8 [5.6%], Brawlhalla [5.6%]
Visual Novel	108	4543	3	Umamusume: Pretty Derby [51.5%], Blue Archive [12.9%], Limbus Company [7.4%], Date Everything! [5.9%], The Hundred Line: Last Defense Academy [4.1%]
Point And Click	109	1525	1	Spirit City: Lofi Sessions [48.8%], The Roottrees Are Dead [5.2%], Escape Simulator [3.3%], Danganronpa V3: Killing Harmony [3%], Unpacking [2.6%]
Total (unique, deduplicated)	3721	233190	100	

Table A3.2. Playtime descriptives by genre. A game's playtime is attributed in full to each of its IGDB labels, so rows are not mutually exclusive and n games exceeds the total unique titles. n games = unique titles; Total hours = playtime attributed to that genre; Mean min/day = average minutes per person-day; Top 5 titles = most-played games by total minutes, with share of genre playtime in brackets (Xbox titles anonymised). The Total row uses deduplicated playtime and distinct titles.

Quantity	Primary	Adjusted
H1a: Within-person total playtime	0.003 [−0.008, 0.014]	0.003 [−0.008, 0.014]
H1b: Between-person total playtime	−0.017 [−0.035, 0.001]	−0.023 [−0.041, −0.006]
H2a: Genre SD (within)	0.032 [0.020, 0.047]	0.032 [0.020, 0.047]
H2b: Genre SD (between)	0.076 [0.044, 0.123]	0.066 [0.037, 0.104]

Table A4.1. Primary versus demographic-adjusted estimates. H1 rows compare M0's total-playtime coefficients with the same coefficients after adding age, gender, and ethnicity. H2 rows compare M1's genre heterogeneity statistic (SD of the 14 genre coefficients, computed at each posterior draw) with its adjusted counterpart. Estimates in SWEMWBS item-mean points per hour/day.

A4 Appendix D: Sensitivity and Supplementary Analyses

This appendix reports the three sensitivity analyses summarised under Robustness in the main text: demographic covariate adjustment, hyperprior-scale sensitivity for the genre heterogeneity parameters, and multiple imputation of wave nonresponse (with distributional diagnostics). It then reports the predictive model comparison (whether genre terms improve out-of-sample prediction), followed by a description of the pre-registered frequentist analysis supplement.

A4.1 Demographic Covariates

The pre-registration specified a sensitivity analysis adjusting the primary models for demographic covariates. We refit M0 (total playtime) and M1 (genre fixed effects) with age, gender, and ethnicity added as population-level predictors, using the subset of participants with complete demographic data.

The adjusted models use 6,464 observations from 1,694 participants with complete demographic data. Table A4.1 compares the primary and covariate-adjusted estimates. The H1 coefficients shift by at most 0.007 points after adjustment, and the ROPE conclusions are unchanged: 100% of the adjusted within-person posterior falls inside the ROPE (primary: 100%), and 100% of the adjusted between-person posterior (primary: 100%). One qualification: the adjusted between-person credible interval ([−0.041, −0.006]) excludes zero, unlike the unadjusted interval; this small negative association remains far inside the ROPE and is consistent with the selection account discussed under H1. The genre heterogeneity estimates are likewise essentially unaffected. Both adjusted models converged cleanly (max R-hat = 1.008, 0 divergent transitions). Demographic composition therefore does not account for the observed associations.

A4.2 Prior Sensitivity

The heterogeneity hyperpriors are the component of the specification where prior choice could most plausibly shape the H2 conclusion, since the Half-Normal(0, 0.1) prior on M2's τ places its own median near the SESOI. As committed to in Deviation D4, we therefore refit M2 with the hyperprior scale halved and doubled (0.05, 0.20) and M3 with its σ_s scale halved and doubled (0.15, 0.60), using otherwise identical specifications (R/03_prior_sensitivity.R).

Table A4.2 shows that the heterogeneity conclusions are not an artefact of the hyperprior scales. Both parameters are essentially unchanged when their hyperprior scales are halved or doubled: M2's τ estimates agree to the third decimal across the fourfold range of scales, and M3's σ_s posterior median and its posterior mass below the SESOI shift only trivially. This is consistent with both posteriors being strongly data-dominated (the posterior SDs are roughly a tenth of the corresponding prior SDs). Across all specifications and scales, the qualitative conclusion is the same as in the main analysis: genre heterogeneity at or below the smallest effect size of interest.

A4.3 Multiple Imputation

The pre-registration specified multiple imputation by chained equations with predictive mean matching. We implemented the two-level grid protocol described in Methods (Missing Data): across the 11,028-cell participant $\times$ wave grid, 4,348 cells (39.4%) had no observed SWEMWBS (Table A4.3), and $m = 100$ imputed datasets were generated. The pilot fractions of missing information for the H1 coefficients were 0.57 (within-person) and 0.91 (between-person): the between-person coefficient depends on person means that are genuinely uncertain for participants with few observed waves, and Rubin's rules propagate that uncertainty into the pooled intervals. Von Hippel's rule would require 167 imputations for a 5%-stable standard error; with $m = 100$ the achieved coefficient of variation of the pooled standard errors is approximately 0.06, which widens the pooled intervals slightly but does not affect the equivalence verdicts.

Table A4.4 compares complete-case and pooled imputed results. The H1 conclusions are unchanged: both pooled 90% confidence intervals fall entirely within the pre-registered equivalence bounds of ± 0.06 , so practical equivalence holds on the imputed data. The between-person interval widens under imputation, reflecting that person-mean wellbeing is uncertain for participants with few observed waves.

Model	Parameter	Hyperprior scale	Posterior median [95% CrI]	P(< SESOI)
M2	τ (between)	0.05	0.018 [0.004, 0.045]	1.00
M2	τ (between)	0.10	0.019 [0.004, 0.050]	0.99
M2	τ (between)	0.20	0.019 [0.003, 0.051]	0.99
M2	τ (within)	0.05	0.013 [0.004, 0.028]	1.00
M2	τ (within)	0.10	0.013 [0.004, 0.028]	1.00
M2	τ (within)	0.20	0.013 [0.004, 0.028]	1.00
M3	σ_s (between)	0.15	0.016 [0.001, 0.044]	1.00
M3	σ_s (between)	0.30	0.016 [0.001, 0.046]	0.99
M3	σ_s (between)	0.60	0.015 [0.001, 0.043]	1.00
M3	σ_s (within)	0.15	0.041 [0.005, 0.099]	0.79
M3	σ_s (within)	0.30	0.041 [0.006, 0.100]	0.78
M3	σ_s (within)	0.60	0.042 [0.006, 0.102]	0.77

Table A4.2. Prior sensitivity for the genre heterogeneity parameters. Each row is a full refit at the hyperprior scale shown; bolded scales are those reported in the main results. P(< SESOI) is the posterior probability that the parameter is below 0.06.

Wave	Possible	Observed	Missing	% missing
1	1838	1835	3	0.2
2	1838	1187	651	35.4
3	1838	1095	743	40.4
4	1838	958	880	47.9
5	1838	871	967	52.6
6	1838	734	1104	60.1

Table A4.3. SWEMWBS missingness by wave on the full participant $\times$ wave grid (analytic sample). Wave 1 was completed on the baseline day; later waves show progressive nonresponse.

For H2, the comparison surfaces the distinction between statistical and practical significance that motivates the SESOI. On the complete cases, the joint Wald test rejects *exact* equality of the 14 within-person genre coefficients ($\chi^2(13) = 25.0$, $p = 0.023$); under the pre-registered decision rule this reading is limited to “at least one pair of genres differs”, with no pairwise interpretation absent formal post-hoc tests. The magnitude of that departure is the quantity the SESOI governs, and it is small: the SD of the 14 coefficient estimates is 0.022 (complete-case) and 0.025 (imputed), both well below 0.06 and consistent with M1’s posterior heterogeneity estimate. On the imputed data the pooled D_1 tests reject neither dimension (within: $p = 0.847$; between: $p = 0.454$), though these tests carry reduced power at this level of missing information (relative increase in variance 0.54 for the within-person constraints). The between-person joint test

Quantity	Complete-case	Imputed (pooled)	Verdict
H1a: Within-person total playtime	0.003 [−0.007, 0.012]	0.006 [−0.008, 0.020]	Practically equivalent
H1b: Between-person total playtime	−0.016 [−0.031, −0.002]	−0.015 [−0.050, 0.021]	Practically equivalent
H2a: Joint test, within-person genre coefficients equal	$\chi^2(13) = 25.0$, $p = 0.023$	$F(13, 10345) = 0.61$, $p = 0.847$	See text
H2b: Joint test, between-person genre coefficients equal	$\chi^2(13) = 13.5$, $p = 0.407$	$F(13, 59737) = 0.99$, $p = 0.454$	Not rejected (either)
Genre SD (within)	0.022	0.025	Below SESOI (both)
Genre SD (between)	0.052	0.055	Below SESOI (both)

Table A4.4. Complete-case versus multiply imputed estimates ($m = 100$; Rubin-pooled with Barnard-Rubin degrees of freedom). H1 rows: total-playtime coefficients from identical frequentist REWB models, with equivalence verdicts applying the pre-registered rule (90% CI within a SESOI of 0.06). H2 rows: joint tests of genre-coefficient equality (complete-case Wald chi-square; imputed pooled D_1 F-test) and the SD of the 14 genre coefficients.

rejects in neither frame. Distributional diagnostics for the imputation are reported in Appendix D (Figure A4.1). In sum, accounting for wave nonresponse leaves every conclusion intact: total playtime is practically equivalent to zero, and genre heterogeneity, statistically detectable or not depending on the frame, remains below the smallest effect size of interest.

A4.3.1 Distributional Diagnostics

The central plausibility check for the imputation model is distributional: imputed wellbeing values should occupy the same range as the observed data and differ from it only in ways explicable by which participants are missing, not by model artefacts.

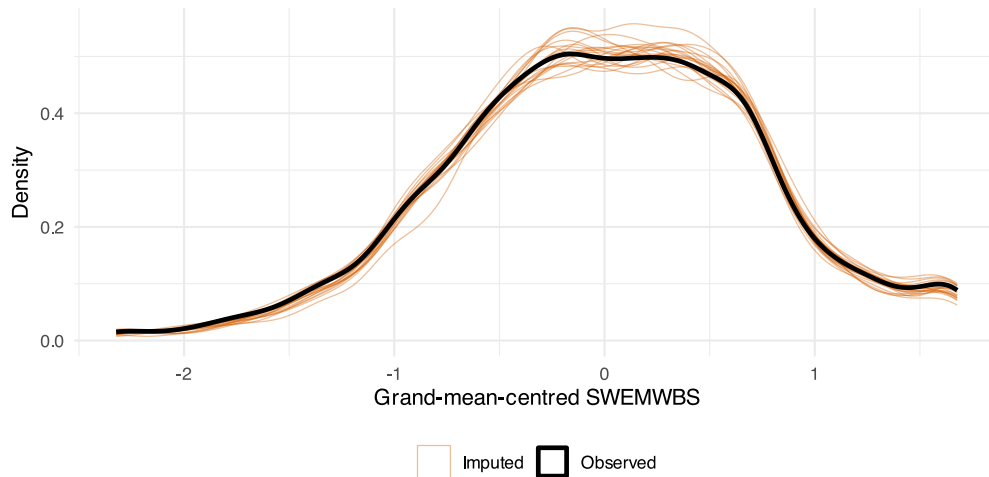


Figure A4.1. Multiple-Imputation Distributional Diagnostic. Observed grand-mean-centred SWEMWBS values (black) against imputed values in 20 of the 100 imputed datasets (thin orange); a subset is shown so the observed curve stays visible, and all 100 behave alike.

Figure A4.1 shows the imputed values to be distributionally almost indistinguishable from the observed data: pooled across all 100 imputations, the imputed values have mean 0.009 and SD 0.74 on the centred scale, against an observed mean of 0 (by construction) and SD of 0.75. The imputation model neither compresses the outcome's variance nor shifts nonresponders' imputed wellbeing to implausible regions of the scale. Convergence traces and strip plots are produced alongside the cached imputation (output/qc_imputation/).

A4.4 Predictive Model Comparison

A complementary way to ask whether genre matters is through predictive performance: does adding genre-specific terms to the total-playtime-only M0 baseline improve out-of-sample prediction? We assess this using approximate leave-one-out cross-validation (LOO-CV) via Pareto-smoothed importance sampling (Vehtari et al., 2017). A positive difference in expected log predictive density (ELPD) relative to M0 indicates that genre terms help.

M0, the model with total playtime only, showed the highest expected log predictive density, and all three genre specifications predicted slightly worse out-of-sample (Table A4.5), with ELPD deficits between one and roughly two standard errors of the difference. Adding genre information, in any of the three specifications, therefore does not improve the prediction of wellbeing over total playtime alone. Pareto- k diagnostics flagged 5.2–5.5% of observations per model ($k > 0.7$; 349–368 observations), so the ELPD values are approximate; this affects the precision of the differences, not the ordering. The inferential reach of this comparison is considered in the Discussion.

A4.5 Pre-Registered Frequentist Analysis

The full pre-registered analysis (primary-genre attribution, frequentist within-between models fit with `lmer`, TOST equivalence tests for H1, joint Wald tests for H2, and multiple imputation via `mice` alongside complete-case results) is reported

Model	dELPD	SE(d)	ELPD	SE	p_loo	LOOIC
M0: Total playtime only	0.0	0.0	-4366.6	78.9	1504.7	8733.1
M3: Multi-membership	-5.7	3.8	-4372.3	79.1	1513.9	8744.6
M1: Fixed effects	-7.6	6.5	-4374.2	78.7	1477.4	8748.3
M2: Hierarchical prior	-10.2	4.9	-4376.8	79.0	1479.6	8753.6

Table A4.5. Leave-one-out cross-validation. Models ordered by expected log predictive density (ELPD); negative dELPD indicates worse out-of-sample prediction than the best-fitting model. SE(d) = standard error of the difference.

in the supplementary materials as a standalone document. It also retains the pre-registered all-platform total playtime measure (including iOS and Android; Appendix B, Deviation D6), providing the comparison between console/PC-only and all-platform H1 estimates.

A4.6 Daily-Timescale Sensitivity Analyses

Four sensitivity refits accompany the exploratory daily analysis (see Exploratory Daily-Timescale Analysis in Methods). Allowing first-order autocorrelated residuals with a within-person day trend (estimated $AR(1) = 0.31$) widens the within-person interval past zero ($\beta = 0.14$, 95% CrI $[-0.02, 0.29]$), as does adjusting for window length ($\beta = 0.15$, 95% CrI $[-0.03, 0.32]$). Excluding the 8% of Xbox sessions recorded with date-only timestamps, whose start times are floored to midnight and whose minutes are spread uniformly over the recorded day, leaves the within-person interval above zero ($\beta = 0.18$, 95% CrI $[0.01, 0.35]$). The all-responses refit, which retains every diary response and scores unobserved play as zero, gives the same between-person lean as the analysed sample ($\beta = -0.52$, 95% CrI $[-1.18, 0.15]$). Two qualifications apply to the daily genre-level intervals: they inherit the exchangeable-residual assumption that the autocorrelation refit relaxes for the total-playtime model only, and the hierarchical-prior model's 18 divergent transitions followed a documented step-size escalation, with estimates stable across both settings.

A5 Appendix E: Hypothesis Test Summaries

This appendix tabulates the posterior summaries and convergence diagnostics underlying the hypothesis tests reported in the Results; the corresponding posterior distributions are shown in Figure 3 (H1) and Figure 4 (H2).

Hypothesis	Model	Parameter	Estimate	95% CrI	% in ROPE	R-hat	Div.
H1: Total playtime, dose-response							
H1a	M0	Within-person slope	0.003	[-0.008, 0.014]	100.0	1.007	0
H1b	M0	Between-person slope	-0.017	[-0.035, 0.001]	100.0	1.007	0
H2a: Genre heterogeneity, within-person							
H2a	M1	SD of genre coefficients	0.032	[0.020, 0.047]	100.0	1.005	0
H2a	M2	τ (hierarchical-prior scale)	0.014	[0.004, 0.028]	100.0	1.003	0
H2a	M3	σ_s (random-slope SD)	0.044	[0.006, 0.100]	78.1	1.003	0
H2b: Genre heterogeneity, between-person							
H2b	M1	SD of genre coefficients	0.076	[0.044, 0.123]	20.9	1.005	0
H2b	M2	τ (hierarchical-prior scale)	0.021	[0.004, 0.050]	99.2	1.003	0
H2b	M3	σ_s (random-slope SD)	0.017	[0.001, 0.046]	99.5	1.003	0

All estimates are in SWEMWBS item-mean points per hour/day. ROPE = region of practical equivalence, -0.06 to +0.06; the heterogeneity parameters are non-negative by construction, so their ROPE spans 0 to 0.06. The H1 slopes are the fixed effects of the standalone total-playtime model (M0). The three H2 statistics quantify genre-to-genre variation in exposure effects: M1 reports the raw SD across the 14 independently estimated fixed-effect coefficients (inflated by sampling noise), M2 the learned hierarchical-prior scale τ on genre fixed effects, and M3 σ_s , the SD of genre random-slope deviations under multi-membership weighting. R-hat is the maximum across all parameters of the fitted model; Div. = number of divergent transitions.

Table A5.1. Main hypothesis tests: posterior estimates, 95% credible intervals, posterior mass inside the ROPE, and convergence diagnostics.

The exploratory daily-timescale analogue of this summary is given in Table A5.2.

Hypothesis	Model	Parameter	Estimate	95% CrI	% in ROPE	R-hat	Div.
H1a-daily	M0	Within-person slope	0.18	[0.00, 0.35]	100.0	1.010	0
H1a-daily	M3	Within-person slope (fixed effect)	0.14	[-0.19, 0.42]	100.0	1.006	0
H1b-daily	M0	Between-person slope	-0.43	[-1.11, 0.27]	95.0	1.010	0
H1b-daily	M3	Between-person slope (fixed effect)	-0.46	[-1.18, 0.27]	93.1	1.006	0
H2a-daily	M1	SD of genre coefficients	0.61	[0.39, 0.85]	99.8	1.011	0
H2a-daily	M2	τ (hierarchical-prior scale)	0.15	[0.01, 0.43]	100.0	1.012	18
H2a-daily	M3	σ_s (random-slope SD)	0.32	[0.01, 0.98]	97.7	1.006	0
H2b-daily	M1	SD of genre coefficients	2.37	[1.48, 3.50]	0.0	1.011	0
H2b-daily	M2	τ (hierarchical-prior scale)	0.84	[0.09, 1.86]	67.3	1.012	18
H2b-daily	M3	σ_s (random-slope SD)	0.19	[0.01, 0.68]	99.5	1.006	0

Table A5.2. Exploratory daily-timescale hypothesis tests (see Exploratory Results: The Daily Timescale): posterior estimates, 95% credible intervals, posterior mass inside the ROPE (± 1 slider point per hour; within-person rows are per hour of same-day play, between-person rows per hour of habitual daily play; heterogeneity parameters are non-negative, so shares are mass below the SESOI of 1), and convergence diagnostics. M1 rows are the SD across the 14 unpooled genre coefficients, computed at each posterior draw, and include estimation error.